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Chemically induced protein dimerization systems

Abstract

The disclosure provides polypeptides, fusion proteins, kits, dimers, nucleic acids, expression vectors, or host cells for use hi chemically induced dimerization systems, exemplified by a chemically induced dimerization system in which two recombinant antibodies dimerize only in the presence of cannabidiol.

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Background/Summary

CROSS REFERENCE (1) This application is a U.S. National Phase of International Application No. PCT/US2020/029052, filed Apr. 21, 2020, which claims priority to U.S. Provisional Application No. 62/837,058, filed Apr. 22, 2019, both of which are incorporated by reference herein in their entirety

REFERENCE TO SEQUENCE LISTING

(1) This application contains a Sequence Listing submitted as an electronic text file named “19-685-PCT_Sequence-Listing_ST25.txt”, having a size in bytes of 19 kb, and created on Apr. 6, 2020. The information contained in this electronic file is hereby incorporated by reference in its entirety pursuant to 37 CFR § 1.52(e)(5).

BACKGROUND

(2) Chemically induced dimerization (CID) systems, in which two proteins dimerize only in the presence of a small molecule ligand, offer versatile tools for small molecule sensing and actuation. However, only a handful of CID systems exist and creating one with the desired sensitivity and specificity for any given ligand is an unsolved problem.

SUMMARY

(3) In one aspect, the disclosure provides recombinant antibodies, comprising a set of complementarity-determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (a)-(aa) in Table 1. In one embodiment, the antibody is a single-chain antibody. In another embodiment, the antibody comprises a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence of SEQ ID NO:82

(4) TABLE-US-00001 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVS AIS-(CDR2)-YYADSVKGRFTISRDN SKNTVYLQMNSLRAEDTATYYC A-(CDR3)-YWGQGTQVTVSS, wherein CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (a)-(aa) in Table 1.

(5) In another embodiment, the antibody comprises a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(6) TABLE-US-00002 (SEQ ID NO: 83) EVX₁LQASGGX₂X₃VQPGGSLRLSX₄AASG-(CDR1)-MGWX₅RQX₆PX₇KEREFVSAIS-(CDR2)-YYX₈DX₉VKGRFTISRDNX₁₀KNTX₁₁YLQM X₁₂SLX₁₃X₁₄EDTAX₁₅YYCX₁₆-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (a)-(aa) in Table 1.

(7) In another embodiment, the set of CDRs comprises a set of complementarity-determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in

any one of rows (a), (i), (n), (u), and (x) in Table 1. In further embodiments, the antibodies comprise one or more functional domains, and/or are bound to a solid support.

(8) In another embodiment, the disclosure provides fusion proteins, comprising: (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3; (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1; and (c) a linker between the first antibody and the second antibody. In one embodiment, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. In another embodiment the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(9) TABLE-US-00003 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVS AIS-(CDR2)-

YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYYC A-(CDR3)-YWGQGTQVTVSS; wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In a further embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(10) TABLE-US-00004 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6PX7 KEREFS AIS-(CDR2)-

YYX8DX9VKGRFTISRDNX10KNTX11YLQM X12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1.

(11) In another embodiment, the disclosure provides kits, comprising: (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3; (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1; and (c) cannabidiol, or a salt thereof. In one embodiment, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. In another embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(12) TABLE-US-00005 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVS AIS-(CDR2)-

YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYYC A-(CDR3)-YWGQGTQVTVSS wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In a further embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%,

98%, 99%, or 100% identical along the length of the amino acid sequence:

(13) TABLE-US-00006 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1.

(14) In another embodiment, the disclosure provides dimers comprising (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3; (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1; and (c) cannabidiol, or a salt thereof, wherein the first recombinant antibody and the second recombinant antibody dimerize in the presence of cannabidiol. In one embodiment, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. In another embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(15) TABLE-US-00007 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVSAIS-(CDR2)-YYADSVKGRFTISRDNNSKNTVYLMNSLRAEDTATYYCA-(CDR3)-YWGQGTQVTVSS; wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In a further embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(16) TABLE-US-00008 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1.

(17) In various other aspects, the disclosure provides nucleic acids encoding the recombinant antibodies or fusion protein of the disclosure, recombinant expression vectors comprising the nucleic acids of the disclosure operatively linked to a control sequence, recombinant host cell comprising the polypeptides, fusion proteins, dimers, nucleic acids and/or the recombinant expression vectors of the disclosure, and uses of the polypeptides, fusion proteins, kits, dimers, nucleic acids, expression vectors, or host cells of the disclosure for any suitable purpose, including but not limited to drug testing, point of care testing, fusion to detectable domains for easy detection, binding to a solid support for simpler visualization, fused to DNA binding proteins and/or transcription factors to control gene regulation, etc.

(18) In another aspect, the disclosure provides selection methods for designing a CID system,

comprising: (a) screening a synthetic combinatorial polypeptide library with a ligand of interest to identify ligand binding polypeptides; (b) screening a synthetic combinatorial polypeptide library with the ligand-bound polypeptide to identify dimerization polypeptides in the library that (i) bind to the ligand binding polypeptide bound to the ligand, and (ii) do not bind to the ligand binding polypeptide in the absence of the ligand.

(19) In one embodiment, the polypeptide library comprises a synthetic combinatorial immunoglobulin, non-immunoglobulin, or computationally designed library, such as a single-domain antibody library. In another embodiment, the screening in step (a) is carried out 1, 2, 3, 4, 5, 6, or more times. In a further embodiment, the screening in step (b) is carried out 1, 2, 3, 4, 5, 6, or more times. In one embodiment, the ligand is 1000 Da, 950 Da, 900 Da, 850 Da, 800 Da, 750 Da, 700 Da, 650 Da, 600 Da, 550 Da, 500 Da, or less in weight. In another embodiment, the polypeptides, fusion proteins, dimers, or CID systems disclosed herein comprise biosensors that can measure the cellular concentration of specific ligands.

Description

DESCRIPTION OF THE FIGURES

- (1) FIG. 1(a)-(c). (a) Ligand-induced dimerization of an anchor binder and a dimerization binder. (b) Schematic of the generation of a synthetic nanobody combinatorial library. (c) Overview of the combinatorial binder library-enabled selection; chemically induced dimerization (COMBINES-CID) method.
- (2) FIG. 2(a)-(b). Anchor binder analysis. (a) Enzyme-linked immunosorbent assay (ELISA) of three anchor binders against biotinylated cannabidiol (CBD) and tetrahydrocannabinol (THC) immobilized on streptavidin-coated plates. Biotin was used as a control. (b) BLI sensorgrams of CA-14 with unlabeled CBD or THC. CA-14 was immobilized on Super Streptavidin biosensors. CBD binding sensorgrams (curves) were modelled using a global fit (lines).
- (3) FIG. 3(a)-(c). CBD-induced nanobody dimerization. (a) BLI sensorgrams of DB-21 and DB-18 immobilized on streptavidin biosensors binding to CA-14 preequilibrated with different concentrations of CBD. (b) SEC analysis of CBD-induced heterodimerization. I) 5 μ M (each) DB-21 and CA-14 and II) 10 μ M (each) DB-18 and CA-14, in the presence or absence of CBD, were crosslinked by 50 μ M bis-N-succinimidyl-(pentaethylene glycol) ester for 30 min at room temperature before SEC. III) 30 μ M non-crosslinked DB-21, DB18, and CA-14 were separated analyzed. Elution volumes of protein standards are marked by triangles. Chromatograms in different groups are shown with different Y scales. (c) Measured $K_{sub.Ds}$ of the ternary complex formation. N.D., not detectable or too weak to be determined.
- (4) FIG. 4(a)-(b). Sandwich ELISA-like detection of CBD. (a) CA-14 was immobilized on a plate and the CBD-induced binding of DB-21 was detected by a horseradish peroxidase-conjugated antibody. (b) The detection of CBD spiked in urine and saliva samples. Limits of detection ($LoDs = \text{mean.sub.blank} + 3 \times (\text{standard deviation of the blank}, n=8)$) for urine and saliva samples were determined to be ~ 0.8 nM (~ 0.25 ng/mL).
- (5) FIG. 5(a)-(c). (a) Design of the nanobody library sequencing with an Illumina NextSeq™ 2 $\times$ 150 bp paired-end kit. Both ends of amplicons were barcoded by 8-bp unique molecular identifiers. (b) Distribution of unique clone counts in merged sequencing reads. (c) Length distribution of CDR3 in sequenced clones.
- (6) FIG. 6(a)-(b). (a) Enrichment of phage titers following each round of biopanning for the anchor binder selection. (b) ELISA of 96 randomly picked clones after the six rounds of the anchor binder selection.
- (7) FIG. 7(a)-(c). (a) Schematic of nanobody expression, purification, and biotinylation. SDS-PAGE analysis of free and streptavidin bead-bound anchor (b) and dimerization (c) binders after

BirA-mediated biotinylation. The biotinylation efficiency was analyzed by incubating biotinylated nanobodies with streptavidin beads and then comparing the ratio of free to bead-bound nanobodies. Bead-bound nanobodies were eluted by boiling beads in SDS sample loading buffer.

(8) FIG. 8(a)-(c). (a) BLI setup for the anchor binder-ligand interaction analysis. (b) BLI sensorgrams showing CBD binding to a panel of anchor binders. CA-14 was done with one sensor per condition, while others are using a single sensor for all conditions. (c) Steady-state binding curve fitting.

(9) FIG. 9(a)-(b). (a) BLI setup for the anchor binder-dimerization binder interaction analysis. (b) BLI sensorgrams showing the anchor binder-dimerization binder interactions in the presence of CBD or THC.

(10) FIG. 10. The sandwich ELISA-like assay of CBD-spiked urine and saliva samples in addition to the samples shown in FIG. 4b. LoD ($\text{mean.sub.blank} + 3 \times (\text{standard deviation, SD.sub.blank})$) for the urine and saliva samples are ~ 0.8 nM.

(11) FIG. 11(a)-(b). Two mechanisms explaining CID. (a) Conformational change induced binding typical of the gibberellin CID system. (b) Three-body binding equilibrium typical of the rapamycin CID system

(12) FIG. 12. Example of a cooperative binding model. Approximating the binding behavior of a BLI experiment where the concentration of immobilized ligand (in this case DB-21) is much smaller than the analyte (in this case CA-14): $[L] \gg [A]$. α was set to 1,000 and $K_{\text{sub.AB}}$ and $K_{\text{sub.BC}}$ was set using BLI data from the CA-14, DB-21 and the dimerization.

(13) FIG. 13. Docking results. (left) The 16 low-energy models of the CA-14 construct. (middle,right) The two putative binding models of CBD suggested from the docking simulation.

DETAILED DESCRIPTION

(14) All references cited are herein incorporated by reference in their entirety. As used herein, the singular forms “a”, “an” and “the” include plural referents unless the context clearly dictates otherwise.

(15) As used herein, the amino acid residues are abbreviated as follows: alanine (Ala; A), asparagine (Asn; N), aspartic acid (Asp; D), arginine (Arg; R), cysteine (Cys; C), glutamic acid (Glu; E), glutamine (Gln; Q), glycine (Gly; G), histidine (His; H), isoleucine (Ile; I), leucine (Leu; L), lysine (Lys; K), methionine (Met; M), phenylalanine (Phe; F), proline (Pro; P), serine (Ser; S), threonine (Thr; T), tryptophan (Trp; W), tyrosine (Tyr; Y), and valine (Val; V).

(16) All embodiments of any aspect of the disclosure can be used in combination, unless the context clearly dictates otherwise.

(17) Unless the context clearly requires otherwise, throughout the description and the claims, the words ‘comprise’, ‘comprising’, and the like are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to”. Words using the singular or plural number also include the plural and singular number, respectively. Additionally, the words “herein,” “above,” and “below” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of the application.

(18) The description of embodiments of the disclosure is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. While the specific embodiments of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize.

(19) In one aspect the disclosure provides recombinant antibodies, comprising a set of complementarity-determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (a)-(aa) in Table 1.

(20) TABLE-US-00009 TABLE 1 CDR sequences of anchor and dimerization binder clones identified in the work. Clones selected for CID construction are highlighted in bold IDs CDR1 CDR2 CDR3 (a) **CA-14** SRSRQYD (SEQ ID

SNQDQPP (SEQ ID FKQHHANGA (SEQ ID NO: 1) NO: 2) NO: 3) (b) CA-17 DTSEDYD (SEQ ID FTSSNHT (SEQ ID FKKHASFP (SEQ ID NO: 4) NO: 5) NO: 6) (c) CA-60 HTSNAYD (SEQ ID SFPDAAV (SEQ ID YKNHPYDPP (SEQ ID NO: 7) NO: 8) NO: 9) (d) DB-1 DTYRLDT (SEQ ID YRTDQDH (SEQ ID GHSWWDLDE (SEQ ID NO: 10) NO: 11) NO: 12) (e) DB-2 TGWEIES (SEQ ID FRANRFE (SEQ ID STFDSPSR (SEQ ID NO: 13) NO: 14) NO: 15) (f) DB-3 YTSFQYV (SEQ ID WLNGQVH (SEQ ID SMVFDHPQSGGGVET NO: 16) NO: 17) (SEQ ID NO: 18) (g) DB-4 YGSDLDS (SEQ ID YAQDDWV (SEQ ID MSIWPEQHH (SEQ ID NO: 19) NO: 20) NO: 21) (h) DB-5 DSSWWDG (SEQ ID WAFDNWR (SEQ ID YTNIDFQAYQSWFQNPPE NO: 22) NO: 23) (SEQ ID NO: 24) (i) **DB-6** RFSWGEE (SEQ ID WAATPWQ (SEQ ID DEWHIGHVS (SEQ ID NO: 25) NO: 26) NO: 27) (j) DB-7 YTSDQDA (SEQ ID SSQSEIA (SEQ ID YRQSVHPQIASM (SEQ NO: 28) NO: 29) ID NO: 30) (k) DB-8 FTFSQEE (SEQ ID FEDGMGK (SEQ ID WWYESHPQFQHQ (SEQ NO: 31) NO: 32) ID NO: 33) (l) DB-9 DTFDLA (SEQ ID WRDNPFR (SEQ ID MLQLHHHDG (SEQ ID NO: 34) NO: 35) NO: 36) (m) DB-10 DTYNWDV (SEQ ID YEPSMYT (SEQ ID MMSSLHTFWANFQSD NO: 37) NO: 38) (SEQ ID NO: 39) (n) **DB-11** TTSDNDT (SEQ ID WNGGRDE (SEQ ID YQDNRSWQE (SEQ ID NO: 40) NO: 41) NO: 42) (o) DB-12 GSYSWDA (SEQ ID YFGHNAY (SEQ ID VHFVKLLNE (SEQ ID NO: 43) NO: 44) NO: 45) (p) DB-13 STYEWYS (SEQ ID WDEDNWN (SEQ ID EPQDGWTGV (SEQ ID NO: 46) NO: 47) NO: 48) (q) DB-14 YTSAGEI (SEQ ID WWDGFAL (SEQ ID AHPSTKMS (SEQ ID NO: 49) NO: 50) NO: 51) (r) DB-15 RFSWGEE (SEQ ID WATAPWQ (SEQ ID YEWHIGHVS (SEQ ID NO: 52) NO: 53) NO: 54) (s) DB-16 DFSSWDA (SEQ ID EGHSMTA (SEQ ID DIEFDLSMNHMYLVQ NO: 55) NO: 56) (SEQ ID NO: 57) (t) DB-17 TTSDWYD (SEQ ID WWPTRAV (SEQ ID DWSFGMMQQ (SEQ ID NO: 58) NO: 59) NO: 60) (u) **DB-18** GYSRADD (SEQ ID FGETDSF (SEQ ID YHNYTNMFE (SEQ ID NO: 61) NO: 62) NO: 63) (v) DB-19 DFYKLYS (SEQ ID WEAGMSH (SEQ ID LQDWMREWE (SEQ ID NO: 64) NO: 65) NO: 66) (w) DB-20 RFSWGEEK (SEQ ID WAAAPWQ (SEQ ID DEWRIDHVS (SEQ ID NO: 67) NO: 68) NO: 69) (x) **DB-21** TTYGQTN (SEQ ID GLQGRDL (SEQ ID FHDFLRMWE (SEQ ID NO: 70) NO: 71) NO: 72) (y) DB-22 DTSNAST (SEQ ID WSSSPGN (SEQ ID MDAFHPQAW (SEQ ID NO: 73) NO: 74) NO: 75) (z) DB-23 YGSFLDS (SEQ ID YAKDDGV (SEQ ID MSIWAEQHH (SEQ ID NO: 76) NO: 77) NO: 78) (aa) DB-24 DYSSTEI (SEQ ID AQPGVQQ (SEQ ID NVARFRHNHD (SEQ ID NO: 79) NO: 80) NO: 81)

(21) As described in the attached appendices, the polypeptides disclosed herein can be used, for example, as part of a chemically induced dimerization (CID) system, in which two recombinant antibodies dimerize only in the presence of cannabidiol.

(22) As disclosed herein, “antibody” refers to immunoglobulin molecules and immunologically active portions of immunoglobulin molecules, i.e., molecules that contain the one of the listed CRD sets and binds to cannabidiol, as described in detail in the attached appendices. As such, the term antibody encompasses not only whole antibody molecules, but also antibody fragments as well as variants (including derivatives) of antibodies and antibody fragments. Such antibody or antibody fragments thereof may include, but are not limited to monoclonal antibodies, humanized antibodies, chimeric antibodies, Fab’, F(ab’).sub.2, Fab, Fv, rIgG, recombinant single chain Fv fragments (scFv), single-chain antibodies (nanobodies), bivalent or bispecific molecules, diabodies, triabodies, and tetrabodies. In one specific embodiment, the antibody comprises a single-chain antibody.

(23) The recombinant antibodies may comprise a suitable scaffold sequence by which to present

the 3 CDRs. Any scaffold sequence may be used, as deemed suitable for an intended use. In one embodiment, the antibody comprises a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(24) TABLE-US-00010 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVS AIS-(CDR2)-

YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYYC A-(CDR3)-YWGQGTQVTVSS

(25) In another embodiment, the antibody comprises a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(26) TABLE-US-00011 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A.

(27) In each of these embodiments, the CDR1, CDR2, and CDR3 may be a set of any of those provided in Table 1 (i.e.: the combination CDR1+CDR2+CDR3 from any single row). The defined sequences comprise a camelid-based universal scaffold, with the three variable complementarity-determining regions (CDRs) interspersed (see FIG. 1b).

(28) In one embodiment, the percent identity requirement is based only on the scaffold sequence (i.e.: not including identity with the CDR sequence). In another embodiment, the percent identity requirement is based on the scaffold sequence and a set of CDRs.

(29) The polypeptides can tolerate significant substitutions, particularly in the scaffold residues. In some embodiments, a given amino acid can be replaced by a residue having similar physiochemical characteristics, e.g., substituting one aliphatic residue for another (such as Ile, Val, Leu, or Ala for one another), or substitution of one polar residue for another (such as between Lys and Arg; Glu and Asp; or Gln and Asn). Other such conservative substitutions, e.g., substitutions of entire regions having similar hydrophobicity characteristics, are known. Amino acids can be grouped according to similarities in the properties of their side chains (in A. L. Lehninger, in Biochemistry, second ed., pp. 73-75, Worth Publishers, New York (1975)): (1) non-polar: Ala (A), Val (V), Leu (L), Ile (I), Pro (P), Phe (F), Trp (W), Met (M); (2) uncharged polar: Gly (G), Ser (S), Thr (T), Cys (C), Tyr (Y), Asn (N), Gln (Q); (3) acidic: Asp (D), Glu (E); (4) basic: Lys (K), Arg (R), His (H). Alternatively, naturally occurring residues can be divided into groups based on common side-chain properties: (1) hydrophobic: Norleucine, Met, Ala, Val, Leu, Ile; (2) neutral hydrophilic: Cys, Ser, Thr, Asn, Gln; (3) acidic: Asp, Glu; (4) basic: His, Lys, Arg; (5) residues that influence chain orientation: Gly, Pro; (6) aromatic: Trp, Tyr, Phe. Non-conservative substitutions will entail exchanging a member of one of these classes for another class. Particular conservative substitutions include, for example; Ala into Gly or into Ser; Arg into Lys; Asn into Gln or into H is; Asp into Glu; Cys into Ser; Gln into Asn; Glu into Asp; Gly into Ala or into Pro; His into Asn or into Gln; Ile into Leu or into Val; Leu into Ile or into Val; Lys into Arg, into Gln or into Glu; Met into Leu, into Tyr or into Ile; Phe into Met, into Leu or into Tyr; Ser into Thr; Thr into Ser; Trp into Tyr; Tyr into Trp; and/or Phe into Val, into Ile or into Leu.

(30) In all of these embodiments, the percent identity requirement does not include any additional functional domain that may be incorporated in the polypeptide.

(31) In one embodiment of any of the above embodiments, the set of CDRs comprises a set of complementarity-determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (a), (i), (n), (u), and (x) in Table 1. These

specific embodiments are shown in Table 2.

(32) TABLE-US-00012 TABLE 2 IDs CDR1 CDR2 CDR3 **(a) CA-14** STSRQYD SNQDQPP FKQHHANGA (SEQ ID NO: 1) (SEQ ID (SEQ ID NO: 3) NO: 2) **(i) DB-6** RFSWGEE WAATPWQ DEWHIGHVS (SEQ ID NO: 25) (SEQ ID (SEQ ID NO: 27) NO: 26) **(n) DB-11** TTSDNDT WNGGRDE YQDNRSWQE (SEQ ID NO: 40) (SEQ ID (SEQ ID NO: 42) NO: 41) **(u) DB-18** GYSRADD FGETDSF YHNYTNMFE (SEQ ID NO: 61) (SEQ ID (SEQ ID NO: 63) NO: 62) **(x) DB-21** TTYGQTN GLQGRDL FHDFLRMWE (SEQ ID NO: 70) (SEQ ID (SEQ ID NO: 72) NO: 71)

(33) In another embodiment of any of the above embodiments, the antibodies further comprise one or more functional domains.

(34) As used herein, a “functional domain” is any polypeptide of interest that might be fused or covalently bound to the polypeptides of the disclosure. In non-limiting embodiments, such functional domains may comprise one or more polypeptide antigens, polypeptide therapeutics, enzymes, detectable domains (ex: fluorescent proteins or fragments thereof), DNA binding proteins, transcription factors, etc. The one or more functional domains may be fused at any appropriate regions within the polypeptides of the disclosure, including but not limited to at the N-terminus or at the C-terminus of the polypeptide.

(35) In a further embodiment, the recombinant antibodies are bound to a solid support. Any suitable solid support may be used, including but not limited to paper, nitrocellulose, beads, cell culture plates, nanoparticles, etc.

(36) In another aspect, the disclosure provides fusion proteins, comprising: (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO: 1, 2, and 3;

(37) TABLE-US-00013 IDs CDR1 CDR2 CDR3 **(a) CA-14** STSRQYD SNQDQPP FKQHHANGA (SEQ ID NO: 1) (SEQ ID (SEQ ID NO: 3) NO: 2) (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1 (reproduced below in relevant part as Table 3):

(38) TABLE-US-00014 TABLE 3 (d) DB-1 DTYRLDT YRTDQDH GHSWWDLDE (SEQ ID NO: 10) (SEQ ID (SEQ ID NO: 12) NO: 11) (e) DB-2 TGWEIES FRANRFE STFDSPSRR (SEQ ID NO: 13) (SEQ ID (SEQ ID NO: 15) NO: 14) (f) DB-3 YTSFQYV WLNGQVH SMVFDHPQSGGGVET (SEQ ID NO: 16) (SEQ ID (SEQ ID NO: 18) NO: 17) (g) DB-4 YGSDLDS YAQDDWV MSIWPEQHH (SEQ ID NO: 19) (SEQ ID (SEQ ID NO: 21) NO: 20) (h) DB-5 DSSWWDG WAFDNWR YTNIDFQAYQSWFQN (SEQ ID NO: 22) (SEQ ID PPE NO: 23) (SEQ ID NO: 24) **(i) DB-6** RFSWGEE WAATPWQ DEWHIGHVS (SEQ ID NO: 25) (SEQ ID (SEQ ID NO: 27) NO: 26) (j) DB-7 YTSDQDA SSQSEIA YRQSVHPQIASM (SEQ ID NO: 28) (SEQ ID (SEQ ID NO: 30) NO: 29) (k) DB-8 FTFSQEE FEDGMKG WWYESHPQFQHQ (SEQ ID NO: 31) (SEQ ID (SEQ ID NO: 33) NO: 32) (l) DB-9 DTFDLA WRDNPFR MLQLHHHDG (SEQ ID NO: 34) (SEQ ID (SEQ ID NO: 36) NO: 35) (m) DB-10 DTYNWDV YEPSMYT MMSSLHTFWANFQSD (SEQ ID NO: 37) (SEQ ID (SEQ ID NO: 39) NO: 38) **(n) DB-11** TTSDNDT WNGGRDE YQDNRSWQE (SEQ ID NO: 40) (SEQ ID (SEQ ID NO: 42) NO: 41) (o) DB-12 GSYSWDA YFGHNAY VHFVKLLNE (SEQ ID NO: 43) (SEQ ID (SEQ ID NO: 45) NO: 44) (p) DB-13 STYEWYS WDEDNWN EPQDGWTGV (SEQ ID NO: 46) (SEQ ID (SEQ ID NO: 48) NO: 47) (q) DB-14 YTSAGEI WWDGFAL AHPSTKMS (SEQ ID NO: 49) (SEQ ID (SEQ ID NO: 51) NO: 50) (r) DB-15 RFSWGEE WATAPWQ YEWHIGHVS (SEQ ID NO: 52) (SEQ ID (SEQ ID NO: 54) NO: 53) (s) DB-16 DFSSWDA EGHSM TA DIEDLSMNHMYLVQ (SEQ ID NO: 55) (SEQ ID (SEQ ID NO: 57) NO: 56) (t) DB-17 TTSDWYD WWPTRAV DWSFGMMQQ (SEQ

ID NO: 58)(SEQ ID (SEQ ID NO: 60) NO: 59) **(u) DB-18** GYSRADD FGETDSF YHNYTNMFE (SEQ ID NO: 61)(SEQ ID (SEQ ID NO: 63) NO: 62) **(v) DB-19** DFYKLYS WEAGMSH LQDWMREWE (SEQ ID NO: 64)(SEQ ID (SEQ ID NO: 66) NO: 65) **(w) DB-20** RFSWGEK WAAAPWQ DEWRIDHVS (SEQ ID NO: 67) (SEQ ID (SEQ ID NO: 69) NO: 68) **(x) DB-21** TTYGQTN GLQGRDL FHDFLRMWE (SEQ ID NO: 70)(SEQ ID (SEQ ID NO: 72) NO: 71) **(y) DB-22** DTSNAST WSSSPGN MDAFHPQAW (SEQ ID NO: 73)(SEQ ID (SEQ ID NO: 75) NO: 74) **(z) DB-23** YGSFLDS YAKDDGV MSIWAEQHH (SEQ ID NO: 76)(SEQ ID (SEQ ID NO: 78) NO: 77) **(aa) DB-24** DYSSTEI AQPQVQQ NVAFRHNHD (SEQ ID NO: 79)(SEQ ID (SEQ ID NO: 81) NO: 80)

and (c) a linker between the first antibody and the second antibody.

(39) In this embodiment, the fusion protein comprises a fused chemically-induced dimerization (CID) system, in which two recombinant antibodies dimerize only in the presence of cannabidiol.

(40) The linker may be of any suitable length and amino acid composition. In various non-limiting embodiment, the linkers may be between 1-200, 2-200, 3-175, 3-150, 3-125, 3-100, 3-75, 3-50, 3-40, 3-30, 3-25, 3-20, 3-15, or 3-10 amino acids in length.

(41) In one embodiment, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. These specific embodiments are shown in Table 4.

(42) TABLE-US-00015 TABLE 4 IDs CDR1 CDR2 CDR3 **(i) DB-6** RFSWGEE WAATPWQ DEWHIGHVS (SEQ ID NO: 25)(SEQ ID (SEQ ID NO: 27) NO: 26) **(n) DB-11** TTSDNDT WNGGRDE YQDNRSWQE (SEQ ID NO: 40)(SEQ ID (SEQ ID NO: 42) NO: 41) **(u) DB-18** GYSRADD FGETDSF YHNYTNMFE (SEQ ID NO: 61) (SEQ ID (SEQ ID NO: 63) NO: 62) **(x) DB-21** TTYGQTN GLQGRDL FHDFLRMWE (SEQ ID NO: 70)(SEQ ID (SEQ ID NO: 72) NO: 71)

(43) In one embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(44) TABLE-US-00016 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREF VSAIS-(CDR2)-YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTAT YYCA-(CDR3)-YWGQGTQVTVSS wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1.

(45) In another embodiment, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(46) TABLE-US-00017 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAXX15YYCX16-(CDR3)-YWGQGTQVTVSS; wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A. wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1.

(47) In each of these embodiments, the CDR1, CDR2, and CDR3 may be a set of any of those provided in Table 1, 2, 3, or 4 (i.e.: the combination CDR1+CDR2+CDR3 from any single row).

The defined sequences comprise a camelid-based universal scaffold, with the three variable complementarity-determining regions (CDRs) interspersed (see FIG. 1b).

(48) In one embodiment, the percent identity requirement is based only on the scaffold sequence (i.e.: not including identity with the CDR sequence). In another embodiment, the percent identity requirement is based on the scaffold sequence and a set of CDRs as recited.

(49) In another aspect, the disclosure provides kit, comprising: (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3;

(50) TABLE-US-00018 IDs CDR1 CDR2 CDR3 (a) CA-14 STSRQYD SNQDQPP

FKQHHANGA (SEQ ID NO: 1) (SEQ ID (SEQ ID NO: 3) NO: 2) (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1 (reproduced above in relevant part as Table 3); and (c) cannabidiol, or a salt thereof.

(51) As described in the attached appendices, the kits disclosed herein can be used, for example, as a chemically induced dimerization (CID) system, in which two recombinant antibodies dimerize only in the presence of cannabidiol.

(52) In one embodiment of the kits, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. These specific embodiments are shown in Table 4.

(53) In one embodiment of the kits, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(54) TABLE-US-00019 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREF VSAIS-(CDR2)-

YYADSVKGRFTISRDN SKNTVYLQMNSLRAEDTAT YYCA-(CDR3)-YWGQGTQVTVSS wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In another embodiment of the kits, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(55) TABLE-US-00020 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A. wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In each of these embodiments, the CDR1, CDR2, and CDR3 may be a set of any of those provided in Table 1, 2, 3, or 4 (i.e.: the combination CDR1+CDR2+CDR3 from any single row). The defined sequences comprise a camelid-based universal scaffold, with the three variable complementarity-determining regions (CDRs) interspersed (see FIG. 1b).

(56) In one embodiment of the kits, the percent identity requirement is based only on the scaffold sequence (i.e.: not including identity with the CDR sequence). In another embodiment, the percent identity requirement is based on the scaffold sequence and a set of CDRs.

(57) In another aspect, the disclosure provides dimers comprising: (a) a first recombinant antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3;

(58) TABLE-US-00021 IDs CDR1 CDR2 CDR3 (a) CA-14 STSRQYD SNQDQPP

FKQHHANGA (SEQ ID NO: 1) (SEQ ID NO: 3) NO: 2) (b) a second recombinant antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1 (reproduced above in relevant part as Table 3); and (c) cannabidiol, or a salt thereof, wherein the first recombinant antibody and the second recombinant antibody dimerize in the presence of cannabidiol.

(59) In one embodiment of the dimers, the second recombinant antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination in any one of rows (i), (n), (u), and (x) in Table 1. These specific embodiments are shown in Table 4.

(60) In one embodiment of the dimers, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(61) TABLE-US-00022 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREF VSAIS-(CDR2)-

YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTAT YYCA-(CDR3)-YWGQGTQVTVSS wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In another embodiment of the dimer, the first antibody and/or the second antibody comprise a scaffold comprising an amino acid sequence at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identical along the length of the amino acid sequence:

(62) TABLE-US-00023 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS;

wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q; X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A. wherein for the second antibody CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination in any one of rows (d)-(aa) in Table 1. In each of these embodiments, the CDR1, CDR2, and CDR3 may be a set of any of those provided in Table 1, 2, 3, or 4 (i.e.: the combination CDR1+CDR2+CDR3 from any single row). The defined sequences comprise a camelid-based universal scaffold, with the three variable complementarity-determining regions (CDRs) interspersed (see FIG. 1b).

(63) In one embodiment of the dimers, the percent identity requirement is based only on the scaffold sequence (i.e.: not including identity with the CDR sequence). In another embodiment, the percent identity requirement is based on the scaffold sequence and a set of CDRs.

(64) The polypeptides and antibodies of the disclosure may include additional residues at the N-terminus, C-terminus, internal to the polypeptide, or a combination thereof; these additional residues are not included in determining the percent identity of the polypeptides of the invention relative to the reference polypeptide. Such residues may be any residues suitable for an intended use, including but not limited to detectable proteins or fragments thereof (also referred to as “tags”). As used herein, “tags” include general detectable moieties (i.e.: fluorescent proteins, antibody epitope tags, etc.), therapeutic agents, purification tags (His tags, etc.), linkers, ligands suitable for purposes of purification, ligands to drive localization of the polypeptide, peptide domains that add functionality to the polypeptides, etc.

(65) As used throughout the present application, the term “polypeptide”, “peptide”, and “protein” are used interchangeably in their broadest sense to refer to a sequence of subunit amino acids of any length, which can include genetically coded and non-genetically coded amino acids,

chemically or biochemically modified or derivatized amino acids, and polypeptides having modified peptide backbones. The polypeptides of the invention may comprise L-amino acids+glycine, D-amino acids+glycine (which are resistant to L-amino acid-specific proteases *in vivo*), or a combination of D- and L-amino acids+glycine. The polypeptides described herein may be chemically synthesized or recombinantly expressed. The polypeptides may be linked to other compounds to promote an increased half-life *in vivo*, such as by PEGylation, HESylation, PASylation, glycosylation, or may be produced as an Fc-fusion or in deimmunized variants. Such linkage can be covalent or non-covalent as is understood by those of skill in the art.

(66) In another aspect, the disclosure provides nucleic acids encoding the recombinant antibody or fusion protein of any embodiment or combination of embodiments of the disclosure. The nucleic acid sequence may comprise single stranded or double stranded RNA or DNA in genomic or cDNA form, or DNA-RNA hybrids, each of which may include chemically or biochemically modified, non-natural, or derivatized nucleotide bases. Such nucleic acid sequences may comprise additional sequences useful for promoting expression and/or purification of the encoded polypeptide, including but not limited to polyA sequences, modified Kozak sequences, and sequences encoding epitope tags, export signals, and secretory signals, nuclear localization signals, and plasma membrane localization signals. It will be apparent to those of skill in the art, based on the teachings herein, what nucleic acid sequences will encode the polypeptides of the disclosure.

(67) In another aspect, the disclosure provides recombinant expression vector comprising the nucleic acid of the disclosure operatively linked to a control sequence. "Expression vector" includes vectors that operatively link a nucleic acid coding region or gene to any control sequences capable of effecting expression of the gene product. "Control sequences" operatively linked to the nucleic acid sequences of the disclosure are nucleic acid sequences capable of effecting the expression of the nucleic acid molecules. The control sequences need not be contiguous with the nucleic acid sequences, so long as they function to direct the expression thereof. Thus, for example, intervening untranslated yet transcribed sequences can be present between a promoter sequence and the nucleic acid sequences and the promoter sequence can still be considered "operably linked" to the coding sequence. Other such control sequences include, but are not limited to, polyadenylation signals, termination signals, and ribosome binding sites. Such expression vectors can be of any type, including but not limited plasmid and viral-based expression vectors. The control sequence used to drive expression of the disclosed nucleic acid sequences in a mammalian system may be constitutive (driven by any of a variety of promoters, including but not limited to, CMV, SV40, RSV, actin, EF) or inducible (driven by any of a number of inducible promoters including, but not limited to, tetracycline, ecdysone, steroid-responsive). The expression vector must be replicable in the host organisms either as an episome or by integration into host chromosomal DNA. In various embodiments, the expression vector may comprise a plasmid, viral-based vector, or any other suitable expression vector.

(68) In a further aspect, the disclosure provides host cells that comprise the nucleic acids or expression vectors (i.e.: episomal or chromosomally integrated) disclosed herein, wherein the host cells can be either prokaryotic or eukaryotic. The cells can be transiently or stably engineered to incorporate the expression vector of the disclosure, using techniques including but not limited to bacterial transformations, calcium phosphate co-precipitation, electroporation, or liposome mediated-, DEAE dextran mediated-, polycationic mediated-, or viral mediated transfection.

(69) In one aspect, the disclosure provides methods for use of the polypeptides, fusion proteins, kits, dimers, nucleic acid, expression vector, or host cell of any embodiment or combination of embodiments disclosed herein for any suitable purpose, including but not limited to drug testing, point of care testing, fusion to detectable domains for easy detection, binding to a solid support for simpler visualization, fused to DNA binding proteins and/or transcription factors to control gene regulation, etc. As shown in the examples that follow, the inventors have demonstrated that CID systems induced by cannabidiol with high ligand selectivity, and which have sensitivity (in a

sandwich ELISA-like assay) of cannabidiol in body fluids with a detection limit of ~0.25 ng/mL. Thus, in one embodiment, the methods are used to detect cannabidiol in body fluids from a subject (such as a human subject).

(70) In another aspect, the disclosure provides methods for making chemically induced dimerization (CID) systems, where two polypeptides only dimerize in the presence of a small-molecule ligand. The methods are described in detail in the examples which follow. Such CID systems may be constructed using two polypeptides selected from a vastly diverse synthetic combinatorial library. The combinatorial library may comprise DNA sequences chemically synthesized by a combinatorial method, such as a trinucleotide mutagenesis technology (PMID: 7838712), to obtain the diversity higher than 10^{10} . The DNA sequences may encode polypeptides consisting of a consensus sequence, typically providing a structural scaffold, such as an immunoglobulin, non-immunoglobulin (PMID: 25931178), or computationally designed scaffold, and variable loop sequences, similar to complementarity determining regions (CDRs) of antibodies, which are structurally flexible and thus can be rationally randomized by incorporating a combination of amino acids under a given ratio into each loop sequence positions. Since both the polypeptides that form a CID system can be selected from a combinatorial library, this selection method may be applicable to generate CID systems for a wide range of small-molecule ligands. These methods are a significant improvement over prior methods which rely on the using existing polypeptide-ligand complexes as a target to search for a dimerization polypeptide, and thus have limited choices of ligands.

(71) The dimerization specificity, i.e., the two polypeptides only form a dimer in the presence of a ligand, may be enhanced by using the flexible loops as the binding site, where the ligand binding triggers a conformational change recognized by a second polypeptide.

(72) Without the conformation change, the second polypeptide cannot dimerize with the first polypeptide. In one embodiment, the methods comprise (a) screening a synthetic combinatorial polypeptide library with a ligand of interest to identify ligand binding polypeptides; (b) screening a synthetic combinatorial polypeptide library with the ligand-bound polypeptide to identify dimerization polypeptides in the library that (i) bind to the ligand binding polypeptide bound to the ligand, and (ii) do not bind to the ligand binding polypeptide in the absence of the ligand. In one embodiment, the polypeptide library may comprise a synthetic combinatorial immunoglobulin, non-immunoglobulin, or computationally designed library, such as a single-domain antibody library. In another embodiment, the screening in step (a) is carried out 1, 2, 3, 4, 5, 6, or more times. In a further embodiment, the screening in step (b) is carried out 1, 2, 3, 4, 5, 6, or more times. In one embodiment, the ligand is 1000 Da, 950 Da, 900 Da, 850 Da, 800 Da, 750 Da, 700 Da, 650 Da, 600 Da, 550 Da, 500 Da, or less in weight. The methods may further comprise any other suitable steps, including but not limited to any step or combination of steps disclosed in the examples that follow.

(73) In one embodiment of the uses and/or CID systems disclosed herein, the polypeptides, fusion proteins, dimers, or CID systems comprise biosensors that can measure the cellular concentration of specific ligands.

EXAMPLES

(74) Chemically induced dimerization (CID) systems, in which two proteins dimerize only in the presence of a small molecule ligand, offer versatile tools for small molecule sensing and actuation. However, only a handful of CID systems exist and creating one with the desired sensitivity and specificity for any given ligand is an unsolved problem. Here, we developed a combinatorial binders-enabled selection of CID (COMBINES-CID) method broadly applicable to different ligands. We demonstrated a proof-of-principle by generating a nanobody-based heterodimerization systems induced by cannabidiol with high ligand selectivity. We applied the CID system to a sensitive sandwich ELISA-like assay of cannabidiol in body fluids with a detection limit of ~0.25 ng/mL. COMBINES-CID provides an efficient, cost-effective solution for expanding the biosensor

toolkit for small-molecule detection.

(75) Creating CID systems for new ligands is an unsolved problem. Here we propose a COMBINES-CID method to select CID proteins for any given ligand—an ‘anchor binder’ that first binds to a ligand, and a ‘dimerization binder’ that only binds to the anchor binder—ligand complex not the unbound anchor binder (FIG. 1a). This method is based on the in vitro selection of vastly diverse protein binder libraries, such as combinatorial antibody libraries,^{sup.16} which can be selected against virtually any epitope. In this work, we focus on a single-domain antibody (or nanobody), a 12-15 kDa functional antibody fragment from camelid comprising a universal scaffold and three variable complementarity-determining regions (CDRs) (FIG. 1b). We reasoned that the three CDR loops might form a binding pocket with adaptable sizes for small-molecule epitopes. Of note, unlike a rigid binding site, the flexible CDR loops might undergo conformational changes upon the ligand binding, providing a basis for the selection of conformationally selective binders only recognizing ligand-bound anchor binders. A stepwise phage-display screening strategy was devised to first obtain anchor binders which were then used as baits to select dimerization binders (FIG. 1c).

(76) As a proof-of-principle, cannabidiol (CBD), a non-psychoactive phytocannabinoid with many medical uses,^{sup.22} was chosen as the ligand. Unlike large, polar, or charged ligands that might be easier targets for binder selection, CBD is hydrophobic and smaller than most ligands in all existing CID systems. Thus, CBD provides a rigorous test of our method.

(77) To enhance the success of selection, we prepared a high-quality nanobody library with high protein diversity and stability. A combinatorial gene library, designed with a thermally stable nanobody scaffold and three rationally randomized CDRs, was chemically synthesized by a trinucleotide mutagenesis technology,^{sup.25} similarly as previously described.^{sup.26} The synthetic DNA library of ~10^{sup.12} sequences was subcloned and transformed into *Escherichia coli* to produce phage-displayed nanobodies. The quality of the phage library was assessed by Sanger and deep sequencing. Approximately 74% of the clones were found within the designed sequences. 39,289,832 out of 41,458,478 merged 2×150 bp paired-end reads were found to be unique (FIG. 5) and the library diversity was estimated to be 1.23 to 7.14×10^{sup.9} by an empirical Bayesian statistical method.^{sup.27} The amino acid distributions of CDRs were close to the expected ratios (data not shown).

(78) To obtain CBD anchor binders (CA), the library was screened using biotinylated CBD as bait. Phage displayed nanobodies were captured by biotinylated CBD bound to streptavidin-coated magnetic beads and eluted by unlabeled CBD (FIG. 1c). After six rounds of selection, three unique clones were found from 96 randomly picked clones. Single-phage ELISA showed that all had specific binding to CBD—biotin—streptavidin but minimal binding to its structural analog, tetrahydrocannabinol (THC)—biotin—streptavidin (FIGS. 2a and 6). The binding selectivity suggests that the minor structural difference in CBD (i.e., the open ring with a phenolic hydroxyl group) is key to the molecular recognition. Interestingly, CDRs in the three nanobodies have little or no sequence similarity (Table 1).

(79) To confirm that the nanobody binding only requires CBD, and not the linker between CBD and biotin or the streptavidin, we measured the binding affinity using unlabeled CBD. Anchor-binder hits were expressed in *E. coli* periplasm and purified by nickel-affinity and size-exclusion chromatography (SEC). Purified nanobodies were enzymatically biotinylated (FIG. 7a-b) and immobilized on streptavidin biosensors to quantify binding affinities to CBD or THC by Bio-Layer Interferometry (BLI) (FIG. 8a). All hits showed high specificity to CBD and dissociation constants (K_Ds) were measured to be 6.0 (CA-14), 26.4 (CA-17), and 3.5 (CA-60) μM (FIGS. 2b and 8b and 8c).

(80) To select specific ligand-induced dimerization, the library was subjected to negative and positive screening using CBD-free and bound CA-14 as baits, respectively (FIG. 1c). After four rounds of selection, 24 unique clones out of 384 were identified. All these DBs showed CBD

concentration-dependent binding to CA-14 by titration ELISA (data not shown). They also showed minimal or no binding to CA-14 in the absence of CBD. Four clones with diverse CDR sequences (Table 1) and high protein expression yields (>2.5 mg per liter of culture) were purified for further characterization.

(81) With the purified DB nanobodies in hand, we first sought to determine their binding affinities to the CA-14-CBD. CBD might directly bind to selected dimerization binders, which can affect the affinity measurement. To rule out this possibility, the four nanobodies were biotinylated (FIG. 7c) and immobilized on streptavidin biosensors, and the interaction with CBD was detected by BLI. Only minimal binding between CBD and the dimerization binders was detected (data not shown). Moreover, the binding-dissociation dynamics of the CA-14-CBD complex can also affect the measurement of dimerization binder affinities. To address this, the affinity was measured by a BLI assay in which the biosensor-immobilized dimerization binders were interacted with CA-14 pre-equilibrated with varied concentrations of CBD (FIG. 9a). The concentrations of the CA-14-CBD complex in equilibrium were calculated based on input CA-14 and CBD concentrations. The CA-14-CBD complex concentration was approximated to be constant because only a minimal fraction of the complex bound to the biosensor tips during the BLI assay. It was confirmed that CA-14 only interacts with the dimerization binders in the presence of CBD (FIG. 3a). The KDs of the four dimerization binders varied from 56.4 nM to 19 μ M (Table 5). CA-14 and DB-21 appear to form the most stable CID complex due to a relatively slow dissociation rate ($k_{\text{sub.off}}=6.17 \times 10^{-4}$ /s; Table 5). All binders showed a high ligand specificity and the dimerization was not induced by THC (FIG. 9b).

(82) TABLE-US-00024

TABLE 5	Dissociation constants of anchor and dimerization binders in the presence or absence of CBD.
Dimerization	K _{sub.D} K _{sub.on} K _{sub.off} binder ID CBD (10 ^{sup.} -9 M) (10 ^{sup.} 5 M ^{sup.} -1s ^{sup.} -1) (10 ^{sup.} -4s ^{sup.} -1)
DB-21 +	56.4 0.11 6.17 - N.D. N.D. N.D.
DB-18 +	553 0.41 227 - N.D. N.D. N.D.
DB-11 +	1,380 0.61 840 - N.D. N.D. N.D.
DB-6 +	19,000 0.096 1,820 - N.D. N.D. N.D.

(83) Next, we confirmed the CBD-induced heterodimerization of the two most stable CID systems (CA-14-DB-21 and CA-14-DB-18) by analytical SEC. As expected, the anchor and dimerization binders by themselves were detected exclusively as monomers, but after adding CBD, SEC peaks corresponding to the heterodimers were observed (FIG. 3b). These results demonstrate a nanobody-based CID binding mechanism (FIG. 3c), which can be explained by a mathematic model of the three-component binding equilibrium (see below). Our model suggests that dimerization binders stabilize the anchor binder-ligand complex by cooperative binding. Furthermore, Rosetta™-based ligand docking calculations shed in-sight onto the nature of the anchor binder-ligand binding (see below).

(84) Lastly, to demonstrate the application in in vitro small molecule detection, we applied the CA-14-DB-21 dimerization system to a sandwich ELISA-like assay (FIG. 4a). Traditional ELISA-based small molecule detection relies on the use of ligand-binding antibodies, which are difficult to obtain by animal immunization because small molecules, by themselves, are non-immunogenic and can only elicit antibodies upon conjugation to protein carriers. Moreover, ELISA requires labeling of ligands—i.e., chemical linking of ligands or their competitive inhibitors to solid supports or reporter molecules—and thus, are not generally applicable because some modifications are difficult to chemically synthesize or can affect ligand-binding activity. However, the sandwich ELISA-like detection only requires the labeling of CID proteins. By analyzing CBD-spiked urine and saliva samples, the assay showed a broad detection range with Limit of Detection (LoD) of ~0.8 nM or ~0.25 ng/mL (FIGS. 4b and 10), which meets requirements for the diagnostic applications..sup.31

(85) In summary, COMBINES-CID enables the efficient search of a vast protein-protein interactome space for ligand-modulated interactions. We demonstrated for the first time that stable CID systems can be constructed with synthetic nanobodies. Our method is applicable to select protein binders with other immunoglobulin, non-immunoglobulin, or computationally designed

scaffolds. The selection is cost-effective and fast since the same combinatorial library can be used for different ligands and high-quality binders were often obtained without in vitro affinity maturation. Together, our results show that this method can generate CID systems for a challenging ligand like CBD. This demonstrates that COMBINES-CID is a suitable method to address the unmet challenge of generating biosensors for highly demanding applications such as point-of-care testing and metabolic engineering.

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Supplemental Methods

(87) 1. Chemical Synthesis

(88) Biotinylated tetrahydrocannabinol (THC) (8) and cannabidiol (CBD) (13) were provided by

the Institute for Protein Design at the University of Washington and were synthesized by DermaXon LLC as described below.

(89) All solvents were reagent grade and used without further purification. All procedures were carried out at room temperature (r.t.) unless otherwise stated. All glassware was oven-dried at 60° C. prior to use. Magnesium sulfate was used as the drying agent. Yields refer to chromatographically pure compounds as determined by TLC or HPLC analysis. The identities and purities of all final products were checked by HPLC-ESI-MS, ¹H, and ¹³C (including APT and DEPT) experiments. NMR spectra were obtained on a Bruker Ascend TM-400 instrument. All spectra were baseline-corrected. All ¹³C NMR spectra were obtained with complete proton decoupling. The signals in the ¹³C NMR spectra were assigned by means of DEPT (Distortionless Enhancement by Polarization Transfer): these methods enable differentiation between the resonances of quaternary carbons (C) and the carbons of CH, CH₂ and CH₃ groups. NMR results of all target compounds were consistent with the assigned structures. High Resolution mass spectrometry (HRMS) analyses were performed on a Waters/Micromass LCT-TOF instrument.

(90) ##STR00001##

Protocols adapted for synthesis of compound 4 Δ⁸-THC from a previous report.¹

(91) 5-pentyl-2-[(1R,5S)-4,6,6-trimethylbicyclo[3.1.1]hept-3-en-2-yl]benzene-1,3-diol (3). To a solution of olivetol 2 (1.98 g, 11 mmol) and p-toluenesulfonic acid (191 mg, 1.1 mmol) in CHCl₃ (65 mL) at r.t. under N₂ was added (s)-cis-verbenol 1 (1.7 g, 11 mmol) in CHCl₃ (30 mL). After being stirred at r.t. for 2.5 h, the reaction mixture was poured into saturated NaHCO₃ and extracted two times with CH₂Cl₂. The combined organic extracts were dried over MgSO₄ and evaporated. The residue was purified by chromatography on silica gel, eluting with EtOAc/hexane (1:9) to afford 3 as a yellow oil (1.34 g, 4.26 mmol, 39%). ¹H NMR (400 MHz, CDCl₃): δ 6.20 (s, 1H), 5.70 (s, 1H), 3.92 (s, 1H), 2.48-2.39 (m, 1H), 2.29 (ddd, J=16.9, 11.5, 5.8 Hz, 1H), 2.18 (t, J=5.5 Hz, 1H), 1.85 (s, 2H), 1.61-1.52 (m, 3H), 1.43 (s, 5H), 1.32 (s, 5H), 0.96 (s, 2H), 0.89 (t, J=6.9 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃): Δ⁸-THC (4). To a solution of compound 3 (0.6 g, 1.9 mmol) in CH₂Cl₂ (35 mL) at 0° C. under N₂ was added BF₃/Et₂O (481 uL, 3.8 mmol), and the mixture was allowed to warm to r.t. After being stirred for 3 h, the mixture was poured into saturated NaHCO₃ and extracted two times with CH₂Cl₂. The combined organic extracts were dried over MgSO₄ and evaporated. The residue was purified by chromatography on silica gel, eluting with CH₂Cl₂/cyclohexane (5:5), to yield 4 as a yellow oil (280 mg, 0.89 mmol, 47%). ¹H NMR (400 MHz, CDCl₃): δ 6.28 (s, 1H), 6.08 (d, J=1.4 Hz, 1H), 5.41 (d, J=4.1 Hz, 1H), 5.18 (d, J=4.9 Hz, 1H), 3.21 (dd, J=16.4, 4.0 Hz, 1H), 2.78-2.61 (m, 1H), 2.40 (dd, J=8.3, 5.8 Hz, 2H), 2.13 (dd, J=10.9, 4.9 Hz, 1H), 1.93-1.73 (m, 3H), 1.68 (s, 3H), 1.61-1.45 (m, 2H), 1.39-1.34 (m, 3H), 1.33-1.19 (m, 4H), 1.09 (s, 3H), 0.87 (t, J=6.9 Hz, 3H).

(92) ##STR00002##

(93) Methyl 2-{[(6aR,10aR)-6,6,9-trimethyl-3-pentyl-6H,6aH,7H,10H,10aH-benzo[c]isochromen-1-yl]oxy}acetate (5). To a solution of compound 4 (280 mg, 0.89 mmol) and potassium carbonate (123 mg, 8.9 mmol) in MEK (10 mL) at r.t. under N₂ was added methyl bromoacetate (163 mg, 1.07 mmol). After being stirred at 80° C. for 18 h, an additional 30 μL of methyl bromoacetate were added since the conversion of the starting material was only approximately 60%. The reaction was stirred at 80° C. for 8 h. 87 mg of cesium carbonate were added and the reaction was stirred at 80° C. overnight. The reaction mixture was poured into saturated NH₄Cl and extracted two times with AcOEt. The combined organic extracts were washed with water, dried over MgSO₄ and evaporated to afford 5 as a yellow oil (344 mg, 0.89 mmol, quantitative yield). ¹H NMR (400 MHz, CDCl₃): δ 6.35 (d, J=1.1 Hz, 1H), 6.11 (d, J=1.1 Hz, 1H), 5.42 (d, J=4.3 Hz, 1H), 4.62 (s, 2H), 3.92-3.75 (m, 3H), 3.34 (dd, J=16.4, 4.5 Hz, 1H), 2.74 (td, J=11.0, 5.0 Hz, 1H), 2.48

(td, J=7.3, 2.1 Hz, 2H), 2.23-2.07 (m, 1H), 1.92-1.75 (m, 3H), 1.71 (s, 3H), 1.64-1.49 (m, 2H), 1.37 (s, 3H), 1.34-1.23 (m, 2H), 1.10 (s, 3H), 0.88 (t, J=6.9 Hz, 4H). ^{sup}.¹³C NMR (100 MHz, CDCl₃): δ 169.46, 157.07, 154.66, 142.53, 135.09, 119.07, 112.53, 111.32, 103.78, 65.51, 52.13, 44.95, 36.08, 35.88, 31.75, 31.56, 30.70, 27.94, 27.57, 26.92, 23.59, 22.55, 18.47, 14.03.

(94) 2-{[(6aR,10aR)-6,6,9-trimethyl-3-pentyl-6H,6aH,7H,10H,10aH-benzo[c]isochromen-1-yl]oxy}acetic acid (6). A solution of compound 5 (269 mg, 0.73 mmol) and lithium hydroxide monohydrate (135 mg, 3.2 mmol) in a mixture of THE (15 mL) and water (250 μL) under N₂ was stirred at 50° C. for 18 h. 250 μL of water were added since the conversion of the starting material was not achieved. The reaction was stirred at 65° C. for 8 h and at r.t. overnight. The reaction mixture was poured into 1 N HCl, and extracted two times with AcOEt. The combined organic extracts were washed with water, dried over MgSO₄ and evaporated. The residue was purified by chromatography on silica gel, eluting with a gradient of CH₂Cl₂/MeOH (2% to 20% of MeOH), to yield 6 as a yellow oil (185 mg, 0.5 mmol, 68%). ^{sup}.¹H NMR (400 MHz, CDCl₃): δ 6.37 (d, J=1.1 Hz, 1H), 6.14 (d, J=1.1 Hz, 1H), 5.42 (d, J=3.6 Hz, 1H), 4.67 (s, 2H), 3.28 (d, J=16.6 Hz, 1H), 2.74 (td, J=11.0, 4.9 Hz, 1H), 2.57-2.42 (m, 2H), 2.22-2.08 (m, 1H), 1.89-1.76 (m, 3H), 1.69 (s, 3H), 1.63-1.52 (m, 2H), 1.38 (s, 3H), 1.35-1.27 (m, 4H), 1.09 (s, 3H), 0.88 (t, J=6.9 Hz, 3H). ^{sup}.¹³C NMR (101 MHz, CDCl₃) δ 156.78, 154.76, 142.67, 135.02, 119.15, 112.58, 111.64, 103.88, 77.35, 77.24, 77.04, 76.72, 65.04, 44.95, 36.15, 35.89, 31.76, 31.58, 30.72, 27.95, 27.57, 23.55, 22.56, 18.48, 14.04.

(95) ##STR00003##

(96) N-(2-{2-[2-(2-{[(6aR,10aR)-6,6,9-trimethyl-3-pentyl-6H,6aH,7H,10H,10aH-benzo[c]isochromen-1-yl]oxy}acetamido)ethoxy]ethoxy}ethyl)-5-{2-oxo-hexahydro-1H-thieno[3,4-d]imidazolidin-4-yl}pentanamide (7). A mixture of O-(7-Azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate HATU (53 mg, 0.14 mmol), DIPEA (28 mg, 0.215 mmol, 0.31 mL) and compound 6 (40 mg, 0.107 mmol) in DCM (2 mL) and DMF (2 mL) was stirred under N₂ at r.t. for 1 h. Biotin-(PEO)₄ amine (50 mg, 0.118 mmol) diluted in a mixture of DCM (2 mL) and DMF (2 mL) was added to the reaction mixture. The reaction mixture was stirred at r.t. for 46 h. The reaction medium was concentrated. Column chromatography (silica gel, gradient MeOH/DCM 2 to 30%) afforded the title compound as a white solid (12 mg, 16%). ^{sup}.¹H NMR (400 MHz, CDCl₃) δ 6.86 (br s, 1H), 6.41 (t, J=5.4 Hz, 1H), 6.37 (s, 1H), 6.25 (br s 1H), 6.19 (s, 1H), 5.44 (d, J=4.1 Hz, 1H), 5.34 (s, 1H), 4.58-4.44 (m, 3H), 4.34-4.27 (m, 1H), 3.64-3.49 (m, 10H), 3.45-3.37 (m, 2H), 3.18-3.10 (m, 1H), 3.07-2.98 (m, 1H), 2.90 (dd, J=12.8, 4.90 Hz, 1H), 2.77-2.66 (m, 2H), 2.47 (dd, J=8.7, 6.4 Hz, 2H), 2.26-2.12 (m, 3H), 1.92-1.60 (m, 10H), 1.62-1.51 (m, 2H), 1.44 (dd, J=15.4, 7.8 Hz, 2H), 1.38 (s, 3H), 1.36-1.24 (m, 4H), 1.10 (s, 3H), 0.88 (t, J=6.84 Hz, 3H). ^{sup}.¹³C NMR (100 MHz, CDCl₃): δ 14.03, 18.44, 22.54, 23.64, 25.53, 27.53, 27.94, 28.09, 28.15, 30.74, 31.55, 31.83, 35.85, 35.92, 36.70, 38.84, 39.12, 40.51, 45.09, 53.43, 55.46, 60.17, 61.81, 67.89, 69.68, 69.97, 70.17, 70.20, 104.59, 111.75, 112.17, 119.70, 134.36, 143.03, 154.68, 156.63, 163.70, 168.92, 173.19, 192.65. HRMS (positive mode) for C₃₉H₆₁N₄O₇S.^{sup}.+[M+1].^{sup}.+ calc 729.4261, found 729.4268.

(97) ##STR00004##

(98) N-[2-(2-{2-[2-(2-{[(6aR,10aR)-6,6,9-trimethyl-3-pentyl-6H,6aH,7H,10H,10aH-benzo[c]isochromen-1-yl]oxy}acetamido)ethoxy]ethoxy}ethoxy)ethyl)-5-{2-oxo-hexahydro-1H-thieno[3,4-d]imidazolidin-4-yl}pentanamide (8). A mixture of O-(7-Azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate HATU (53 mg, 0.14 mmol), DIPEA (28 mg, 0.215 mmol, 0.31 mL) and compound 6 (40 mg, 0.107 mmol) in DCM (4 mL) and DMF (2 mL) was stirred under N₂ at r.t. for 1 h. Biotin-(PEO)₄ amine (50 mg, 0.118 mmol) was added. The reaction mixture was stirred at r.t. for 46 h. The reaction medium was concentrated. Column chromatography (silica gel, gradient MeOH/DCM 2 to 30%) afforded the title compound as a white solid (12 mg, 16%). ^{sup}.¹H NMR (400 MHz, CDCl₃) δ 6.89 (br s, 1H), 6.51 (t, J=5.4 Hz, 1H), 6.37 (d, J=1.3 Hz, 1H), 6.19 (d, J=1.3 Hz, 1H), 6.09 (s, 1H), 5.44 (d, J=4.4 Hz, 1H), 5.16 (s,

1H), 4.65-4.41 (m, 3H), 4.42-4.24 (m, 1H), 3.69-3.49 (m, 13H), 3.42 (dd, J=9.9, 5.1 Hz, 2H), 3.15 (dt, J=11.9, 6.0 Hz, 1H), 3.09-2.95 (m, 1H), 2.90 (dd, J=12.8, 5.0 Hz, 11H), 2.71 (dd, J=17.8, 8.5 Hz, 2H), 2.47 (dd, J=8.7, 6.4 Hz, 2H), 2.34-2.06 (m, 3H), 1.96-1.60 (m, 11H), 1.60-1.49 (m, 2H), 1.45 (dd, J=15.4, 7.8 Hz, 21H), 1.38 (s, 2H), 1.34-1.25 (m, 4H), 1.10 (s, 3H), 0.89 (t, J=6.8 Hz, 3H). ^{sup}.13C NMR (101 MHz, CDCl₃) δ 173.10, 168.84, 163.48, 156.66, 154.66, 143.01, 134.44, 119.64, 112.18, 111.71, 104.60, 70.48, 70.44, 70.32, 70.09, 69.95, 69.71, 67.90, 61.76, 60.13, 55.37, 45.09, 40.52, 39.15, 38.85, 36.69, 35.85, 31.82, 31.55, 30.75, 28.09, 27.95, 27.54, 25.49, 23.64, 22.54, 18.45, 14.04. HRMS (positive mode) for C₄₁H₆₅N₄O₈S.^{sup}.+[M+1].^{sup}.+ calc 773.4523 found 773.4554.

(99) ##STR00005##

(100) Methyl 2-{3-hydroxy-2-[(1R,6R)-3-methyl-6-(prop-1-en-2-yl)cyclohex-3-en-1-yl]-5-pentylphenoxy}acetate (10). A solution of CBD, compound 9 (50 mg, 0.16 mmol) and cesium carbonate (52 mg, 0.16 mmol) in MEK (2 mL) under N₂ was heated at 80° C. After being stirred at 80° C. for 1 h, methyl bromoacetate (15 μL, 0.16 mmol) was added. The reaction was stirred at 80° C. for 8 h. The reaction mixture was concentrated. Two batches of the reaction were combined and were purified by chromatography on silica gel, eluting with a gradient of cyclohexane/ethyl acetate (95/5) to (65/35). A mixture of CBD and the expected product (50/50, according to HPLC quantification) was obtained (140 mg overall).

(101) 2-{3-hydroxy-2-[(1R,6R)-3-methyl-6-(prop-1-en-2-yl)cyclohex-3-en-1-yl]-5-pentylphenoxy}acetic acid (11). A solution of mixture 10 previously obtained (140 mg) and lithium hydroxide monohydrate (65 mg, 1.5 mmol) in a mixture of THF (5 mL) and water (125 μL) under N₂ was stirred at room temperature overnight. The residue was purified by chromatography on silica gel, eluting with a gradient of CH₂Cl₂/MeOH (100/0 to 90/10), to yield 11 as a yellow oil (45 mg, 25% over the two steps). ^{sup}.1H NMR (400 MHz, CDCl₃): δ ppm 6.38 (s, 1H), 6.13 (br. s., 2H), 5.57 (br. s., 1H), 4.62-4.48 (m, 3H), 4.36 (br. s., 1H), 4.08-3.98 (m, 1H), 2.53-2.33 (m, 3H), 2.30-2.16 (m, 1H), 2.14-2.06 (m, 1H), 1.85-1.72 (m, 5H), 1.64 (s, 3H), 1.56 (dt, J=15.00, 7.56 Hz, 2H), 1.37-1.22 (m, 4H), 0.88 (t, J=6.96 Hz, 3H).

(102) ##STR00006##

(103) N-(2-{2-[2-(2-{3-hydroxy-2-[(1R,6R)-3-methyl-6-(prop-1-en-2-yl)cyclohex-2-en-1-yl]-5-pentylphenoxy}acetamido)ethoxy]ethoxy}ethyl)-5-{2-oxo-hexahydro-1H-thieno[3,4-d]imidazolidin-4-yl}pentanamide (12). A mixture of O-(7-Azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate HATU (32 mg, 0.084 mmol), DIPEA (20 μL, 0.128 mmol) and compound 11 (24 mg, 0.064 mmol) in DCM (1 mL) and DMF (1 mL) was stirred under N₂ at r.t. for 1 h. Biotin-(PEO)₃ amine (13 mg, 0.035 mmol) diluted in DMF (1 mL) was added to the reaction mixture. The reaction mixture was stirred at r.t. for 46 h and concentrated. Column chromatography (silica gel, gradient MeOH/DCM 0 to 15%) afforded the title compound as a white solid (9 mg, 35%). ^{sup}.1H NMR (400 MHz, CDCl₃) δ 6.83 (br s, 1H), 6.45-6.34 (m, 1H), 6.25 (m, 11H), 6.13 (br s, 11H), 6.05 (br s, 11H), 5.57 (br s, 1H), 5.31 (br s, 1H), 4.64 (br s, 1H), 4.50-4.56 (m, 1H), 4.50-4.37 (m, 3H), 4.36-4.29 (m, 1H), 3.98-3.89 (m, 1H), 3.64-3.46 (m, 10H), 3.46-3.37 (m, 2H), 3.19-3.12 (m, 1H), 2.92 (dd, J=12.92, 4.89 Hz, 1H), 2.73 (d, J=12.80 Hz, 2H), 2.56-2.35 (m, 3H), 2.22 (t, J=7.22 Hz, 3H), 2.13-2.03 (m, 1H), 1.87-1.60 (m, 11H), 1.54 (dt, J=14.87, 7.50 Hz, 3H), 1.49-1.41 (m, 2H), 1.35-1.21 (m, 4H), 0.87 (t, J=6.90 Hz, 3H). ^{sup}.13C NMR (101 MHz, CDCl₃) δ ppm 14.06, 20.00, 22.54, 25.47, 28.02, 29.87, 30.84, 31.48, 35.89, 38.84, 39.18, 40.40, 40.51, 53.58, 55.35, 60.28, 61.92, 69.89, 70.17, 104.13, 110.85, 111.13, 120.30, 123.55, 129.39, 163.51, 169.35, 173.40. HRMS (positive mode) for C₃₉H₆₁N₄O₇S.^{sup}.+[M+1].^{sup}.+ calc 729.4261, found 729.4260.

(104) ##STR00007##

(105) N-[2-(2-{2-[2-(2-{3-hydroxy-2-[(1R,6R)-3-methyl-6-(prop-1-en-2-yl)cyclohex-2-en-1-yl]-5-pentylphenoxy}acetamido)ethoxy]ethoxy}ethoxy)ethyl]-5-{2-oxo-hexahydro-1H-thieno[3,4-d]imidazolidin-4-yl}pentanamide (13). A mixture of O-(7-Azabenzotriazol-1-yl)-N,N,N',N'-

tetramethyluronium hexafluorophosphate HATU (30 mg, 0.077 mmol), DIPEA (20 μ L, 0.118 mmol) and compound 11 (22 mg, 0.059 mmol) in DCM (1 mL) and DMF (1 mL) was stirred under N.sub.2 at r.t. for 1 h. Biotin-(PEO).sub.4 amine (27 mg, 0.065 mmol) diluted in DCM (1 mL) was added. The reaction mixture was stirred at r.t. for 46 h. The reaction medium was concentrated. Column chromatography (silica gel, gradient MeOH/DCM 2 to 15%) afforded the title compound as a white solid (20 mg, 45%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.83 (br s, 1H), 6.66-6.47 (m, 1H), 6.37 (m, 1H), 6.31-6.17 (m, 1H), 6.12 (br s, 1H), 5.58 (br s, 1H), 5.35 (br s, 1H), 4.63 (br s, 1H), 4.56-4.46 (m, 2H), 4.46-4.38 (m, 2H), 4.38-4.26 (m, 1H), 3.98-3.89 (m, 1H), 3.66-3.46 (m, 14H), 3.46-3.37 (m, 2H), 3.19-3.11 (m, 1H), 2.91 (dd, J=12.92, 4.89 Hz, 1H), 2.74 (d, J=12.80 Hz, 2H), 2.56-2.34 (m, 3H), 2.15-2.31 (m, 3H), 2.13-2.01 (m, 11H), 1.86-1.60 (m, 111H), 1.54 (dt, J=14.96, 7.51 Hz, 3H), 1.48-1.39 (m, 2H), 1.35-1.21 (m, 4H), 0.88 (t, J=6.96 Hz, 3H). .sup.13C NMR (101 MHz, CDCl.sub.3) δ ppm 14.05, 19.92, 22.52, 23.73, 25.47, 27.56, 28.04, 28.06, 29.95, 30.83, 31.47, 35.83, 35.87, 36.40, 38.83, 39.18, 46.03, 53.43, 55.37, 60.23, 61.83, 69.90, 69.98, 70.06, 70.31, 70.36, 70.39, 104.15, 10.92, 110.94, 110.97, 111.04, 111.08, 120.22, 123.76, 123.77, 123.84, 129.16, 163.77 169.17, 173.32. HRMS (positive mode) for C₄₁H₆₅N₄O₈S.sup.+ [M+1].sup.+ calc 773.4523, found 773.4562.

(106) 2. Nanobody Library Construction and Validation

(107) Library construction: The DNA library was designed with a universal nanobody scaffold and randomized CDR sequences similarly to a reported method.² The combinatorial DNA library was chemically synthesized by a trinucleotide mutagenesis technology (Thermo Fisher Scientific). To subclone the library into a phagemid vector, ~300 ng of the synthetic DNA library was used as template to set up 30 $\times$ 50 μ L PCR reactions, each with 1 μ L of Platinum SuperFi DNA polymerase (Thermo Fisher Scientific) and 0.5 μ M (each) primers. The PCR protocol included an initial denaturation step at 98 $^{\circ}$ C. for 30 s followed by 20 cycles of 98 $^{\circ}$ C. for 10 s, 70 $^{\circ}$ C. for 10 s, and 72 $^{\circ}$ C. for 10 s, and a final step extension at 72 $^{\circ}$ C. for 5 min. ~40 μ g purified PCR products and ~60 μ g of a pADL-23c phagemid vector (Antibody Design Labs) were digested for 1 h at 37 $^{\circ}$ C. with BglI (Thermo Fisher Scientific) and purified before the ligation. ~8.5 μ g digested vector and ~3.9 μ g inserts were ligated with 260 units of T4 DNA ligase (Thermo Fisher Scientific) in a 2,600 μ L reaction at 4 $^{\circ}$ C. for overnight. Ligated products were purified by a PurelinkTM PCR quick purification kit (Invitrogen) to obtain a final volume of 50 μ L (~278.9 ng/ μ L) and then transformed to *E. coli* electrocompetent TG1 cells (Lucigen) in 30 electroporation cuvettes, each containing 1.4 μ L ligated products and 50 μ L cells, by following the manufacturer's instruction (1,800 V, 10 μ F, 600 Ω). Electroporated cells were resuspended with a warm recovery medium (Lucigen) and incubated by shaking (250 rpm) at 37 $^{\circ}$ C. for 1 h and then mixed with ~470 mL 2 $\times$ YT, 2% glucose. Cells were plated on 500 2 $\times$ YT-ampicillin-glucose agar plates (140 mm), grown overnight at 30 $^{\circ}$ C., and scraped with 2 $\times$ TY. The library was aliquoted and stored in 25% glycerol at -80 $^{\circ}$ C.

(108) Next generation sequencing library validation: To sequence the phage-displayed nanobody library, phagemid DNAs were isolated from purified phage particles using a QIAprep Spin M13 kit (Qiagen). DNA concentration was measured by a NanodropTM 2000 spectrophotometer (Thermo Fisher Scientific). A two-step low-cycle-number PCR was performed to introduce the Illumina adaptors and 8-bp unique molecular identifiers (UMIs) to the 3' and 5' ends of amplicons with specific primers. The library was sequenced by an Illumina NextSeqTM platform using a 2 $\times$ 150 bp high-output kit. The sequencing data were first processed to trim UMI sequences using FlexbarTM.³ Low-quality reads were filtered with a minimum quality score of 20. To accommodate errors within UMIs while retrieving CDR sequences, undetermined bases in UMIs or shorter barcodes were allowed. Clean reads were then aligned to CDR-adjacent scaffold sequences to extract designed CDR sequences at a 0.1 mismatch rate. Three CDR sequences were obtained by merging paired-end reads. To retain correct sequences, CDR sequences were required to be in the same reading frame without any undetermined base or stop codon.

(109) A custom Perl script, shown below, was used to analyze translated protein sequences. CDR

protein sequence logos were generated with WebLogo3.6..sup.4

```
(110) TABLE-US-00025 #! /usr/bin/perl -w use strict ; my $fastqL = $ARGV [0] ; my $fastqR =
$ARGV [1] ; my %processedReads ; my %phagereads ; open (FQL, "$fastqL") || die; while (my
$1n = <FQL> ) { if ($.%4 !=1) (next ; ) chomp $1n; my @code = split(/\./, $1n) ; my
$coord = $code [0] ; my $tagcodeL = $code [1] ; my $cdr1_rna = $code [2] ; my $cdr2_rna
= $code [3] ; #remove the peptides containing undertermined nu1 if ($cdr1_rna =~ /N/ or
$cdr2_rna =~ /N/ ) { next; } # filter for fixed peptide length if (length ($cdr1_rna) %
3 ==0 and length ($cdr2_rna) %3 ==0) { $phagereads{$coord} = $tagcodeL. ' | '. $cdr1_rna. ' |
'. $cdr2_rna ; } } close (FQL) ; open (FQR, "$fastqR") || die; my $headertag; while (my $1n =
<FQR> ) { chomp $1n; if {$.%4 ==1}{ $headertag = $1n; next ; } if {$.%
4 ==2){ my @code = split(/\./, $headertag) ; my $coord = $code [0] ; $coord =~ s/
2\ : N\ : 0\ : / 1\ : N\ : 0\ : / ; my $tagcodeR = $code [1] ; my $cdr3_rna = $1n; #
remove the peptides containing undertermined nu1 if ($cdr3_rna =~ /N/ ) { next; }
# filter for frame shift if (length ($cdr3_rna) % 3 ==0 and exists $phagereads {$coord}) {
$processedReads {$coord} = $phagereads {$coord}. ' | '. $cdr3_rna. ' | '. $tagcodeR ; } }
} close (FQR) ; open ( PRO, ">$ARGV [2] ") || die; my %countPeptide; foreach my $key (keys
%processedReads) { my @data = split{/\./, $processedReads{$key}} ; my $c1 = translate
($data [1] ) ; my $c2 = translate ($data [2] ) ; my $c3 = translate ($data [3] ) ; print PRO
$key, "\t", $data [0] , "\t", $data [4] , "\t", $data [1] , "\t", $data [2] , "\t", $data [3] , "\t", $c1, "\t",
$c2, "\t", $c3, "\n" ; $countPeptide {$data [0] , '_' . $data [4] } {$c1. '_' . $c2. '_' . $c3} ++ ; }
close(PRO); foreach my $key (keys %countPeptide) { foreach my $sky (keys % {$countPeptide
{ $key } } ) { print '>', $key, '-', $countPeptide { $key } { $sky } , "\n" ; print $sky, "\n"
; } } sub reverse_complement { my $dna = shift; # reverse the DNA sequence my
$revcomp = reverse ($dna) ; # complement the reversed DNA sequence $revcomp =~
tr/ATCGN/TAGCN/ ; return $revcomp ; } sub translate { my (%genetic_code) = {
'TCA' => 'S', # Serine 'TCC' => 'S', # Serine 'TCG' => 'S', # Serine
'TCT' => 'S', # Serine 'TTC' => 'F', # Phenylalanine 'TTT' => 'F', #P
henylalanine 'TTA' => 'L', # Leucine 'TTG' => 'L', # Leucine 'TAC' =>
'Y', # Tyrosine 'TAT' => 'Y', # Tyrosine 'TAA' => '4', # Stop 'TAG' => '4',
# Stop 'TGC' => 'C', # Cysteine 'TGT' => 'C', # Cysteine 'TGA' => '4', #
Stop 'TGG' => 'W', # Tryptophan 'CTA' => 'L', # Leucine 'CTC' => 'L', #
Leucine 'CTG' => 'L', # Leucine 'CTT' => 'L', # Leucine 'CCA' => 'P', #
Proline 'CAT' => 'H', # Histidine 'CAA' => 'Q', # Glutamine 'CAG' =>
'Q', # Glutamine 'CGA' => 'R', # Arginine 'CGC' => 'R', # Arginine
'CGG' => 'R', # Arginine 'CGT' => 'R', # Arginine 'ATA' => 'I', #
Isoleucine 'ATC' => 'I', # Isoleucine 'ATT' => 'I', # Isoleucine 'ATG' =>
'M', # Methionine 'ACA' => 'T', # Threonine 'ACC' => 'T', # Threonine
'ACG' => 'T', # Threonine 'ACT' => 'T', # Threonine 'AAC' => 'N', #
Asparagine 'AAT' => 'N', # Asparagine 'AAA' => 'K', # Lysine 'AAG' =>
'K', # Lysine 'AGC' => 'S', # Serine 'AGT' => 'S', # Serine 'AGA' => 'R',
# Arginine 'AGG' => 'R', # Arginine 'CCC' => 'P', # Proline 'CCG' => 'P',
# Proline 'CCT' => 'P', # Proline 'CAC' => 'H', # Histidine 'GTA' => 'V',
# Valine 'GTC' => 'V', # Valine 'GTG' => 'V', # Valine 'GTT' => 'V', #
Valine 'GCA' => 'A', # Alanine 'GCC' => 'A', # Alanine 'GCG' => 'A', #
Alanine 'GCT' => 'A', # Alanine 'GAC' => 'D', # Aspartic Acid 'GAT' =>
'D', # Aspartic Acid 'GAA' => 'E', # Glutamic Acid 'GAG' => 'E', # Glutamic
Acid 'GGA' => 'G', # Glycine 'GGC' => 'G', # Glycine 'GGG' => 'G', #
Glycine 'GGT' => 'G', # Glycine } ; my $rna = shift; # if (length ($rna) %3!
=0) { # return (0); # } my $protein = " " ; for (my $i =0; $i <lenght ($rna)-2; $
i +=3) { my $codon = substr ($rna, $i, 3); if (not exists $genetic_code {$codon
```



```

    }) {
        $protein .= 6;
    } else {
        $protein .= $genetic_code {$scodon};
    }
    }
    }
    return ($protein);
}

```

3. Phage Display Selection

(111) To prepare a phage library, the phagemid-containing TG1 bacterial stock from the “Library construction” was diluted with 2×YT media with 2% glucose and 100 µg/mL ampicillin to OD600 of ~0.1 and cultured at 37° C. to OD600 of ~0.4 to 0.5. Cells were superinfected by adding a helper phage CM13 at 5×10⁹ pfu/mL for 1 h, pelleted to remove the glucose, resuspended with fresh 2×YT media containing 100 µg/mL ampicillin and 50 µg/mL kanamycin, and incubated at 25° C. for overnight. To purify phage particles, cells were removed by centrifugation at 5,000×g, 4° C. for 30 min and phage particles in the supernatant were precipitated with polyethylene glycol (PEG). The supernatant was added with 1/5 volume PEG/NaCl solution (20% (w/v) polyethylene glycol 6,000 and 2.5 M NaCl), placed on ice for 1 h, and centrifuged at 12,000×g, 4° C. for 30 min. Phage pellets were resuspended in 1×PBS and stored at 4° C. for short-term use or with 25% glycerol at -80° C. for long-term storage.

(112) Anchor binder selection was performed using biotin- and biotinylated-CBD-bound streptavidin magnetic beads for negative and positive selections, respectively, in each selection round. Briefly, 300 µL 10 µM biotin or biotinylated CBD were captured by 300 µL streptavidin-coated magnetic beads (Dynabeads™ M-280 Streptavidin, Thermo Fisher Scientific) and blocked with 1% casein and 1% BSA in 1×PBS pH 7.4 at r.t. for 1 h. In each round, the phage-displayed nanobody library was incubated with the biotin-bound beads for 1 h at r.t. to remove off-target binders and the supernatant was incubated with biotinylated CBD-bound beads for 1 h. Beads were washed with 10×0.05% PBST (1×PBS with 0.05% v/v Tween 20) and phage particles were competitively eluted with 50, 10 and 1 µM CBD in the first, second, and third selection rounds, respectively, and 100 nM CBD in the fourth to sixth rounds. After six rounds of biopanning, single colonies were picked and validated by phage ELISA followed by DNA sequencing.

(113) Dimerization binders were selected using CBD-free and bound CA-14 for the negative and positive selections, respectively. Briefly, to prepare the bait for each selection round, 600 µL 1 µM biotinylated CA-14 were captured by 600 µL streptavidin beads and blocked with 1% casein and 1% BSA in 1×PBS pH 7.4 for 1 h at r.t. The beads were divided by a 2:1 ratio for the negative and positive selections, respectively. For the positive selection, the CA-14 bound beads were incubated with 50, 10, 1, and 1 µM CBD to generate the CBD-CA-14 complex as the bait from the rounds 1 to 4, respectively. Biopanning was performed as described above except that bound phages were eluted with 100 mM triethylamine. Four rounds of biopanning were performed before single colony picking and validation.

(114) 4. Phage ELISA

(115) To prepare single-phage ELISA, individual colonies were randomly picked, inoculated into 250 µL media (2×TY, 2% glucose, 100 µg/mL ampicillin) per well in sterile deep-well plates (Thermo Fisher Scientific), and grown at 37° C. for overnight. 10 µL cultures were inoculated into 500 µL fresh media and cells were grown to OD600 of ~0.5 and infected by CM13 helper phage with the multiplicity of infection (MOI) of ~18. The cultures were incubated at 37° C. for 1 h, added with kanamycin of 50 µg/mL final concentration, and grown at 25° C. for overnight. Plates were centrifuged for 10 min at 3,000×g and phage-containing supernatants were transferred to fresh plates for the ELISA assays.

(116) For anchor-binder phage ELISA, ELISA plates (Nunc MaxiSorp™, Thermo Fisher Scientific) were coated with 100 µL 5 g/mL streptavidin in coating buffer (100 mM carbonate buffer, pH 8.6) at 4° C. for overnight. After washing with 3×0.05% PBST (1×PBS with 0.05% v/v Tween 20), each well was added with 100 µL 1 µM biotinylated CBD and incubated at r.t. for 1 h. Wells were washed with 5×0.05% PBST, blocked with 1% casein in 1×PBS, and then added with 100 µL phage supernatants. Phage particles were incubated with CBD in plates at r.t. for 1 h. Wells were washed with 10×0.05% PBST, added with 100 µL HRP-M13 major coat protein antibody

(RL-ph1, Santa Cruz Biotechnology; 1:2000 dilution with 1×PBS with 1% casein), and incubated at r.t. for 1 h. A colorimetric detection was performed using a 1-Step Ultra TMB ELISA substrate solution (Thermo Fisher Scientific) and OD450 was measured with a SpectraMax™ Plus 384 microplate reader (Molecular Devices).

(117) The dimerization-binder phage ELISA used the plate-immobilized anchor binder as the bait. ~100 nM biotinylated anchor binder, CA-14, was bound to a streptavidin-coated plate prepared as above described. The immobilized CA-14 was bound to different concentrations of CBD in 1×PBS, pH 7.4, to form the anchor binder-CBD complex before binding to phage-displayed dimerization binders. Other steps were performed similarly to the anchor-binder phage ELISA.

(118) 5. Protein Expression and Purification

(119) All nanobodies were expressed as a C-terminal AviTagged and His-tagged form in *E. coli* and purified by Ni-affinity and size-exclusion chromatography. In brief, *E. coli* strain WK6 was transformed with the expression constructs and grown in TB medium at 37° C. to an OD600 of ~0.7 before induction with 1 mM isopropyl-β-D-galactopyranoside (IPTG) at 28° C. for overnight.

Harvested cell pellets from 1-liter cultures were resuspended in 15 mL ice-cold TES buffer (0.2 M Tris-HCL pH 8.0, 0.5 mM EDTA, 0.5 M sucrose) and incubated with gently shaking on ice for 1 h. To release proteins from the periplasm by osmotic shock, the resuspended pellets were added with 30 mL of TES/4 buffer (1:4 dilution of the TES buffer in ddH.sub.2O) and gently shaken on ice for 45 min. Cell debris was removed by centrifugation at 15,000×g, 4° C. for 30 mins. The supernatant was loaded onto a 5 mL HisTrap™ column (GE Healthcare) pre-equilibrated with the lysis buffer (50 mM sodium phosphate, pH 8.0, 300 mM NaCl, 10 mM imidazole, 10% glycerol). The column was washed with a washing buffer (50 mM sodium phosphate, pH 8.0, 300 mM NaCl, 20 mM imidazole, 10% glycerol) and then His-tagged proteins were eluted with an elution buffer (50 mM sodium phosphate, pH 8.0, 300 mM NaCl, 250 mM imidazole, 10% glycerol). Eluates were concentrated with an Amicon™ Ultra-15 centrifugal filter unit (3 kDa cutoff, Millipore).

Concentrated proteins were loaded onto a HiLoad™ 16/600 Superdex 200 pg column (GE Healthcare) pre-equilibrated with a storage buffer (1×PBS, 5% glycerol). Eluted proteins were concentrated, examined by SDS-PAGE, and quantified by a Bradford assay (BioRad), then flash frozen in 100 μL aliquots by liquid N.sub.2 and stored at -80° C.

(120) 6. Protein Biotinylation

(121) Nanobodies bearing AviTag were biotinylated by BirA using a BirA-500 kit (Avidity).sup.5 Typically, 90 μL BiomixA (10× concentration: 0.5 M bicine buffer, pH 8.3), 90 μL BiomixB (10× concentration: 100 mM ATP, 100 mM Mg(OAc).sub.2, 500 μM d-biotin), 4 μL 1 mg/mL BirA, and 216 μL ddH.sub.2O were added to 500 μL ~1 mg/mL AviTagged nanobodies to a final volume of 900 μL. The biotinylation mixture was incubated at r.t. for 1 h and then loaded onto a HiPrep 26/10 desalting column (GE Healthcare) pre-equilibrated with a storage buffer (1×PBS, 5% glycerol). Eluted proteins were concentrated, examined by SDS-PAGE, and quantified by the Bradford assay, flash frozen in 100 μL aliquots by liquid N.sub.2, and stored at -80° C.

(122) 7. BioLayer Interferometry

(123) The nanobody binding kinetics was analyzed using an Octet™ RED96 system (ForteBio) and Streptavidin (SA) or Super Streptavidin (SSA) biosensors. For the anchor-binder analysis, 200 nM biotinylated anchor binders were immobilized on SSA biosensors with a binding assay buffer (1×PBS, pH 7.4, 0.05% Tween 20, 0.2% BSA, 3% methanol; Note: because CBD and THC were dissolved in methanol to make stock solutions, 3% methanol was added to the buffer to fairly compare samples and controls). Serial dilutions of CBD and THC were used for the anchor binder assays. Dissociation constants (K.sub.Ds) of anchor binder-ligand interactions were calculated based on a steady-state analysis using Octet™ RED96 data analysis software. For the dimerization-binder analysis, 200 nM biotinylated dimerization binders were immobilized on SA biosensors with the binding assay buffer and then assayed with 1 μM CA-14 pre-equilibrated with serial dilutions of CBD or THC. K.sub.Ds, k.sub.ons, and k.sub.off of the interactions between anchor binder-

ligand complexes and dimerization binders were calculated based on a heterogeneous ligand model global fit (2:1) of the data and simulated binding equilibria of the anchor binder and CBD described in Supplementary Notes.

(124) 8. Analytical SEC

(125) Anchor and dimerization nanobodies were analyzed by SEC with a Superdex™ 75 increase 10/300 GL column (GE Healthcare). For non-crosslinked samples, 500 μ L ~30 μ M nanobodies were injected into the column equilibrated with 1 $\times$ PBS and eluted at a flow rate of 0.75 mL/min at 4° C. The column was calibrated with molecular weight standards (Sigma-Aldrich). Crosslinked samples were prepared by incubating 5 or 10 μ M proteins in the presence or absence of CBD in 1 $\times$ PBS with 0.1 mM BS(PEG).sub.5 at r.t. for 30 mins. 500 μ L crosslinked samples were injected to the column.

(126) 9. ELISA-Like Assay to Determine CBD Detection Sensitivity

(127) Urine and saliva samples were collected from three healthy volunteers (two males and one female). The urine and saliva samples were spun for 3 minutes at 14,000 $\times$ g and the supernatants were diluted by $\frac{1}{5}$ with 1 $\times$ PBS, pH 7.4, and spiked with CBD. Serial dilutions (0 to 1,000 nM CBD) of samples were used for the dimerization binder ELISA similarly to that described above. Each dilution was repeated 8 times and Limit of Detection (LoD) was calculated by $\text{mean.sub.blank} + 3 \times \text{SD.sub.blank}$.

Supplemental Notes

(128) 1. Binding Model Expectations Based on Previous CID Systems and Ligand Docking

(129) Two mechanisms can explain the currently existing CID systems.sup.6 (FIG. 11): a) the chemical inducer works as a bridge between the two proteins dimerizing, and b) the chemical induces a large structural change upon binding one of the proteins, this structural change then allows for dimerization to the other protein.

(130) The gibberellin CID system is an example of protein dimerization which depends on a necessary conformational change in one of the dimerizing proteins. In the gibberellin CID system, the GID1A domain contains an N-terminal extension that, upon binding to gibberellin, forms a stable interaction surface for the DELLA domain..sup.7 In this system gibberellin is completely buried in a deep binding pocket and only the conformationally changed GID1A domain is free to bind the DELLA domain. Such system is kinetically simple with only two dissociation constants defining the system (FIG. 11a).

(131) The rapamycin CID system is an example of the “bridge molecule” mechanism. In this system the protein FKBP binds to one part of the rapamycin molecule, while the protein FRB binds to the other end..sup.8,9 In such systems, because both proteins can bind independently to the small molecule, the kinetics are more complex defined by two dissociation constants and a parameter, α , quantifying “binding cooperation” (FIG. 11b). When $\alpha > 0$, this is termed positive cooperation and represents a binding enhancement upon ternary complex formation. When $\alpha < 0$, this is termed negative cooperation and represents a binding diminishment upon ternary complex formation. Others have studied the three-body binding equilibrium in great detail..sup.10 Using established affinities,.sup.8 the rapamycin CID system has positive cooperation of ~2,000.

(132) We hypothesize that our CBD CID system works by either of the two above described binding mechanisms. We primarily tested CBD anchor 14 (CA-14) which has relatively short CDR sequences (Table 1). With short CDRs it is unlikely that CBD binding induces a gibberellin-like conformational change through CDR loop stabilization. This also makes it unlikely that CBD is completely buried with no exposed parts, although, we cannot rule out this possibility. Indeed, our docking study suggests that CBD is still exposed after binding to CA-14 (FIG. 13). This leads us to believe that CBD induced dimerization is governed by the “bridge molecule” mechanism similar to the rapamycin CID. The distinction between these binding models is important because binding through a bridge molecule leads to a system with possible auto-inhibition,.sup.10 also known as the hook effect or the prozone effect. Auto-inhibition negatively affects dimerization at high

concentrations of the bridge molecule because it competes against dimerization. The effect is described in detail in..sup.10

(133) Bio Layer Interferometry to determine binding kinetics

(134) In order to determine the binding kinetics of our nanobody CID system we performed a series of bio layer interferometry (BLI) experiments. In all experiments we used an Octet RED96 system (ForteBio) and Streptavidin Biosensors (ForteBio). 200 nM biotinylated protein (FIGS. 7b & 4C) were immobilized on biosensors (Super Streptavidin, SSA used for anchor binder; Streptavidin, SA used for dimerization binder) with a binding assay buffer (1×PBS, pH 7.4, 0.05% Tween 20, 0.2% BSA, 3% methanol; Note: because CBD and THC were dissolved in methanol to make stock solutions, 3% methanol was added to the buffer to fairly compare samples and controls). Ligand loading was followed by a quenching step using free biotin. All experiments were performed using four measurements to control for sensor drift and unspecific binding: No ligand, no analyte With ligand, no analyte No ligand, with analyte With ligand, with analyte

(135) The ligand is defined as the protein loaded to the SSA sensor and the analyte is either protein or small molecule in solution. The final sensorgrams were derived from: (d-c)-(b-a)

(136) We first determined the binding affinity between the CBD anchor 14 (CA-14) and CBD via steady-state analysis (FIG. 8). For this experiment, BLI sensorgrams were collected using one SSA sensor per CBD concentration over six concentrations. Data analysis were performed in ForteBio DataAnalysis v9 using a global 1:1 model (standard Langmuir isotherm). The global model requires binding parameters for all six concentrations to be the same, except the ligand loading amount, R.sub.max, which was unlinked to allow for differences in ligand loading caused by sensor variance. Steady-state analysis was then performed using the steady-state values estimated by this global model. We performed a similar experiment, using a single sensor for multiple concentrations, on the dimerization binders (DB). We observed minimal binding between DBs and CBD, but the KDs are too weak to be determined (data not shown).

(137) Next, we wanted to determine the binding affinity between CA-14 bound to CBD (referred to as CA-14*) and a panel of DBs. We observed no detectable binding between CA-14 and any of the DBs (FIG. 3a). We also rely on the observation that the DBs have weak binding to CBD, and therefore we can assume that only CA-14* binds the DB loaded to the sensor. We control the CA-14* concentration by using a fixed concentration of 1 μM CA-14 pre-incubated to equilibrium with six different concentrations of CBD. These CA-14* concentrations were then allowed to bind to DB loaded SA sensors and binding was measured (FIG. 9). One sensor was used per CBD concentration. The sensorgrams initially showed a fast binding phase which was followed by a phase with slow binding. Such biphasic binding is an indication of a heterogeneous surface..sup.11 Attempts to optimize the binding behavior was unsuccessful so we turned towards fitting methods that take into account multiple binding phases. One such method is implemented as a publicly available software called EVILFIT..sup.12 EVILFIT integrates over a grid of possible affinities and assigns probabilities to each. Using CA-14's affinity to CBD, we calculated the CA-14* concentration at equilibrium for the different CBD concentrations and used this as the analyte concentrations to EVILFIT. At the two largest CBD concentrations we observed CBD based auto-inhibition of dimerization (discussed below) and therefore these were removed. When restricting our analysis to the four smallest CBD concentrations, EVILFIT showed good agreement between the data and the fit (data not shown). Using these fits, we located the affinity distribution for the first binding phase and integrated over this to derive a point estimate of the affinities between CA-14 and the DBs (Table 5).

(138) Suggested Binding Model for CA-14 Dimerizing DB-21

(139) CA-14 and DB-21 is our most well characterized binding pair. BLI analysis established the binding affinity between CA-14 and CBD to be 6.0 μM (FIG. 8). Docking studies suggested that CBD bound CA-14 in a way that exposes smaller parts of CBD (FIG. 13) potentially acting as a bridge to DB-21. This hypothesis is supported by the characteristic auto-inhibition we observed in

our ELISA titration data (data not shown), indicating that CBD at high concentrations is competing with dimerization. We observe a similar kind of auto-inhibition at the two highest CBD concentrations in our BLI assay (FIG. 9).

(140) If CBD is acting as a bridging molecule it must also bind DB-21. However, another BLI analysis showed that DB-21 only binds CBD very weakly (data not shown), and therefore bridging cannot by itself explain the strong dimerization we observe (FIG. 9). The observations can, however, be explained by positive cooperation, which enhances dimerization, and has before been shown to cause ~2,000 enhancement in rapamycin inducible dimerization..sup.8

(141) Douglass et al..sup.10 derives the equilibrium solution to such three-body binding systems with cooperation. Unfortunately, we cannot reliably measure DB-21's affinity to CBD ($K_{sub.BC}$ in FIG. 11b) due to its weak affinity and the limited solubility of CBD. However, we have been able to determine CBD's affinity to CA-14 and CA-14*'s affinity to DB-21 ($K_{sub.AB}$ and $K_{sub.BC}/\alpha$ in FIG. 11b). As for the cooperation parameter, α , it must be so large that $K_{sub.BC}$ turns form undetectable by BLI to 56 nM ($K_{sub.BC}/\alpha$). $\alpha=1,000$ is a reasonable guess, also within the limit of what has been observed in the rapamycin CID, making $K_{sub.BC}$ equal to 56 μ M. Plugging this into the three-body binding model from Douglass et al. (FIG. 12), we get an explanation for the auto-inhibition we observe in ELISA titrations, as well as a reason why the two highest CBD concentrations in our BLI sensorgrams of dimerization (FIG. 9) does not follow a standard steady-state curve.

(142) 2. Docking Studies of CBD Binding CA-14

(143) Rosetta™-based protein modelling and protein/ligand docking was carried out with the anchor binder CA-14 and CBD. First, models of the nanobody CA-14 were constructed with RosettaCM™..sup.13 Modelling used four template structures (pdb ids: 1mqk, 1t2j, 1uac, and 6cnw). The command used to run RosettaCM™ follows:

(144) TABLE-US-00026 \$ROSETTA/source/bin/rosetta_scripts.default.linuxgccrelease \ - in:file:fasta CA14.fasta \ -parser: :protocol hybrid.xml \ -relax: :dualspace \ - relax:default_repeats 5 \ -default_max_cycles 200 \ -score: weights beta_cart \ -beta The command above was provided the following XML script:

(145) TABLE-US-00027 <ROSETTASCRIPTS> <SCOREFXNS> <ScoreFunction name="stage1" weights="score3"> <Reweight scoretype="atom_pair_constraint" weight="0.1" / > < / ScoreFunction> <ScoreFunction name ="stage2" weights="score4_smooth_cart"> <Reweight scoretype="atom_pair_constraint" weight="0.1" / > < / ScoreFunction> <ScoreFunction name ="fullatom" weights ="beta_cart"> <Reweight scoretype="atom_pair_constraint" weight="0.1" / > < / ScoreFunction> < / SCOREFXNS> <MOVERS> <Hybridize name ="hybridize" stage1_scorefxn="stage1" stage2_scorefxn="stage2" fa_scorefxn="fullatom" batch="1"> <Template pdb="1mqkH_201.pdb" weight="1.0" cst_file="AUTO" / > <Template pdb="1t2jA_202.pdb" weight="1.0" cst_file="AUTO" / > <Template pdb="6cnwA_203.pdb" weight="1.0" cst_file="AUTO" / > <Template pdb="1uacH_204.pdb" weight="1.0" cst_file="AUTO" / > < / Hybridize> < / MOVERS> <PROTOCOLS> <Add mover="hybridize" / > < / PROTOCOLS> <OUTPUT scorefxn="fullatom" / > < / ROSETTASCRIPTS>

(146) RosettaCM™ was run in parallel, generating a total of 400 initial models. The 16 lowest energy models (FIG. 13) were then used as receptor structures in ligand docking.

(147) Ligand docking used an unpublished docking protocol in Rosetta, with a recently developed generalized energy function using prior methodology.sup.14 for energy function fitting.

(148) A CBD ligand conformation was generated using OpenBabel™..sup.15 This model was then minimized using AM1 and AM1-BCC partial charges were generated with antechamber..sup.16 Finally, the refined small molecule was converted to a Rosetta parameter file with the following command:

(149) TABLE-US-00028 python \$ROSETTA/source/scripts/python/public/generic_potential/mol2genparams.py -amide_chi -s LG1.mol2

Docking in Rosetta™ was carried out running the following command:

(150) TABLE-US-00029 \$ROSETTA/source/bin/rosetta_scripts.linuxgccrelease \ -database ~/Rosetta_efunc/database \ -s input.pdb \ -parser:protocol dock.xml \ -beta_cart \ -extra_res_fa CBD.params \ -no_autogen_cart_improper

The input XML, dock.xml:

(151) TABLE-US-00030 \$ROSETTA/source/bin/rosetta_scripts.linuxgccrelease \ -database ~/Rosetta_efunc/database \ -s input.pdb \ -parser:protocol dock.xml \ -beta_cart \ -extra_res_fa CBD.params \ -no_autogen_cart_improper

The input XML, dock.xml:
<ROSETTASCRIPTS> <SCOREFXNS> <ScoreFunction name="dockscore" weights="beta"> <Reweight scoretype="fa_rep" weight="0.2" /> <Reweight scoretype="coordinate_constraint" weight="0.1" /> </ScoreFunction> <ScoreFunction name="relaxscore" weights="beta_cart" /> </SCOREFXNS> <MOVERS> <GALigandDock name="dock" runmode="dockflex" scorefxn="dockscore" scorefxn_relax="relaxscore" sidechains="auto"> <Stage repeats="10" npool="100" pmut="0.2" smoothing="0.375" rmsdthreshold="2.0" maxiter="50" pack_cycles="100" ramp_schedule="0.1, 1.0"/> </GALigandDock> </MOVERS> <PROTOCOLS> <Add mover="dock" /> </PROTOCOLS> <OUTPUT scorefxn="relaxscore" /> <ROSETTASCRIPTS>

(152) The input files from this run were the ten lowest-energy receptor structures from stage 1. Initially, input ligands were placed in a random orientation such that the ligand center of mass was coincident with the C-beta of residue 33, the residue closest to the center of mass of the variable loops in the nanobody.

(153) The previous XML samples an ensemble of 100 structures in a single trajectory. We ran a total of 160 such trajectories, ten trajectories for each of the 16 initial models, generating a total of 16,000 models. Visual inspection of the lowest-energy 20 docked conformations yielded two unique configurations, illustrated in FIG. 13.

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Claims

1. A single-chain antibody, comprising a set of complementarity-determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination comprising the amino acid sequences: (a) SEQ ID NO:1, SEQ ID NO:2, and SEQ ID NO:3; (b) SEQ ID NO:4, SEQ ID NO:5, and SEQ ID NO: 6; (c) SEQ ID NO:7, SEQ ID NO:8, and SEQ ID NO: 9; (d) SEQ ID NO:10, SEQ ID NO:11, and SEQ ID NO: 12; (e) SEQ ID NO:13, SEQ ID NO:14, and SEQ ID NO: 15; (f) SEQ ID NO:16, SEQ ID NO:17, and SEQ ID NO: 18; (g) SEQ ID NO:19, SEQ ID NO:20, and SEQ ID NO:21; (h) SEQ ID NO:22, SEQ ID NO:23, and SEQ ID NO:24; (i) SEQ ID NO:25, SEQ ID NO:26, and SEQ ID NO:27; (j) SEQ ID NO:28, SEQ ID NO:29, and SEQ ID NO:30; (k) SEQ ID NO:31, SEQ ID NO:32, and SEQ ID NO:33 (l) SEQ ID NO:34, SEQ ID NO:35, and SEQ ID NO:36; (m) SEQ ID NO:37, SEQ ID NO:38, and SEQ ID NO:39; (n) SEQ ID NO:40, SEQ ID NO:41, and SEQ ID NO:42; (o) SEQ ID NO:43, SEQ ID NO:44, and SEQ ID NO:45; (p) SEQ ID NO:46, SEQ ID NO:47, and SEQ ID NO:48; (q) SEQ ID NO:49, SEQ ID NO:50, and SEQ ID NO:51; (r) SEQ ID NO:52, SEQ ID NO:53, and SEQ ID NO:54; (s) SEQ ID NO:55, SEQ ID NO:56, and SEQ ID NO:57; (t) SEQ ID NO:58, SEQ ID NO:59, and SEQ ID NO: 60; (u) SEQ ID NO:61, SEQ ID NO:62, and SEQ ID NO: 63; (v) SEQ ID NO:64, SEQ ID NO:65, and SEQ ID NO:66; (w) SEQ ID NO:67, SEQ ID NO:68, and SEQ ID NO:69; (x) SEQ ID NO:70, SEQ ID NO:71, and SEQ ID NO: 72; (y) SEQ ID NO:73, SEQ ID NO:74, and SEQ ID NO:75; (z) SEQ ID NO:76, SEQ ID NO:77, and SEQ ID NO:78; or (aa) SEQ ID NO:79, SEQ ID NO:80, and SEQ ID NO:81.
2. The single-chain antibody of claim 1, wherein the single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence of SEQ ID NO:82 TABLE-US-00031 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFV SAIS-(CDR2)-YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYY CA-(CDR3)-YWGQGTQVTVSS, wherein CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination according to any one of options (a)-(aa) in claim 1.
3. The single-chain antibody of claim 1, wherein the single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00032 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6P X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YL QMX12SLX13X14EDTX15YYCX16-(CDR3)-YWGQGTQVTVSS; wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q, X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein CDR1, CDR2, and CDR3 comprise a CDR1, CDR2, and CDR3 combination according to any one of options (a)-(aa) in claim 1.
4. The single-chain antibody of claim 1, wherein CDR1 comprises the amino acid sequence of SEQ ID NO:1, CDR2 comprises the amino acid sequence of SEQ ID NO:2, and CDR3 comprises the amino acid sequence of SEQ ID NO:3.
5. The single-chain antibody of claim 1, further comprising one or more functional domains.
6. The single-chain antibody of claim 1, wherein the single-chain antibody is bound to a solid support.
7. A fusion protein, comprising: (a) a first single-chain antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3; (b) a second single-chain antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination according to any one of options (d)-(aa) in claim 1; and (c) a linker between the first single-chain antibody and the second single-chain antibody.

8. The fusion protein of claim 7, wherein the second single-chain antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination comprising the amino acid sequences: (i) SEQ ID NO:70, SEQ ID NO:71, and SEQ ID NO:72; (ii) SEQ ID NO:25, SEQ ID NO:26, and SEQ ID NO:27; (iii) SEQ ID NO:40, SEQ ID NO:41, and SEQ ID NO:42; or (iv) SEQ ID NO:61, SEQ ID NO:62 and SEQ ID NO: 63.

9. A kit, comprising: (a) a first single-chain antibody, comprising the set of complementarity determining regions (CDRs) of SEQ ID NO:1, 2, and 3; (b) a second single-chain antibody, comprising a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 combination according to any one of options (d)-(aa) in claim 1; and (c) cannabidiol, or a salt thereof.

10. The kit of claim 9, wherein the second single-chain antibody comprises a set of complementarity determining regions (CDRs) selected from the group consisting of a CDR1, CDR2, and CDR3 comprising the amino acid sequences: (i) SEQ ID NO:70, SEQ ID NO:71, and SEQ ID NO: 72; (ii) SEQ ID NO:25, SEQ ID NO:26, and SEQ ID NO:27; (iii) SEQ ID NO:40, SEQ ID NO:41, and SEQ ID NO:42; or (iv) SEQ ID NO:61, SEQ ID NO:62 and SEQ ID NO: 63.

11. The kit of claim 9, wherein the first single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00033 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVSAIS-(CDR2)-YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYY CA-(CDR3)-YWGQGTQVTVSS wherein CDR1 comprises the amino acid sequence of SEQ ID NO:1, CDR2 comprises the amino acid sequence of SEQ ID NO:2, and CDR3 comprising the amino acid sequence of SEQ ID NO:3.

12. The kit of claim 11, wherein the second single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00034 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVSAIS-(CDR2)-YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYY CA-(CDR3)-YWGQGTQVTVSS; wherein CDR1, CDR2, and CDR3 comprise the amino acid sequences: (i) SEQ ID NO:70, SEQ ID NO:71, and SEQ ID NO: 72; (ii) SEQ ID NO:25, SEQ ID NO:26, and SEQ ID NO:27; (iii) SEQ ID NO:40, SEQ ID NO:41, and SEQ ID NO:42; or (iv) SEQ ID NO:61, SEQ ID NO:62, and SEQ ID NO: 63.

13. The kit of claim 11, wherein the second single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00035 (SEQ ID NO: 82) EVQLQASGGGFVQPGGSLRLSCAASG-(CDR1)-MGWFRQAPGKEREFVSAIS-(CDR2)-YYADSVKGRFTISRDN SKNTVY LQMNSLRAEDTATYY CA-(CDR3)-YWGQGTQVTVSS; wherein CDR1 comprises the amino acid sequence of SEQ ID NO:70, CDR2 comprises and amino acid sequence of SEQ ID NO:71, and CDR3 comprises the amino acid sequence of SEQ ID NO:72.

14. The kit of claim 9, wherein the first single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00036 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6PX7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11Y LQM X12SLX13X14EDTAX15YYCX16-(CDR3)-YWGQGTQVTVSS; wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q, X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein CDR1 comprises the amino acid sequence of SEQ ID NO:1, CDR2 comprises the

amino acid sequence of SEQ ID NO:2, and CDR3 comprising the amino acid sequence of SEQ ID NO:3.

15. The kit of claim 14, wherein the second single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical to the amino acid sequence: TABLE-US-00037 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6X7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YL QMX12SLX13X14EDTAXX15YYCX16-(CDR3)-YWGQGTQVTVSS; wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q, X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein CDR1, CDR2, and CDR3 comprise the amino acid sequences: (i) SEQ ID NO:70, SEQ ID NO:71, and SEQ ID NO:72; (ii) SEQ ID NO:25, SEQ ID NO:26, and SEQ ID NO:27; (iii) SEQ ID NO:40, SEQ ID NO:41, and SEQ ID NO:42; or (iv) SEQ ID NO:61, SEQ ID NO:62, and SEQ ID NO: 63.

16. The kit of claim 14, wherein the second single-chain antibody comprises a scaffold comprising an amino acid sequence at least 50% identical of to the amino acid sequence: TABLE-US-00038 (SEQ ID NO: 83) EVX1LQASGGX2X3VQPGGSLRLSX4AASG-(CDR1)-MGWX5RQX6PX7KEREFVSAIS-(CDR2)-YYX8DX9VKGRFTISRDNX10KNTX11YLQMX12SLX13X14EDTAXX15YYCX16-(CDR3)-YWGQGTQVTVSS; wherein X1 is K, Q, N, R, D, or E; X2 is D, G, E, A, V, L, or I; X3 is L, F, W, G, A, V, L, or I; X4 is K, R, D, E, N, or Q, X5 is V, P, G, A, L, I, or M; X6 is T, A, S, G, V, L, or I; X7 is E, G, D, A, V, L, or I; X8 is P, A, M, G, V, L, or I; X9 is T, or S; X10 is A, S, M, G, V, L, or I; X11 is L, V, G, A, or I; X12 is S, N, Q, or T; X13 is K, R, D, or E; X14 is S, A, T, G, V, L, or I; X15 is M, T, V, L, or I; X16 is V, G, L, I, or A; and wherein CDR1 comprises the amino acid sequence of SEQ ID NO:70, CDR2 comprises and amino acid sequence of SEQ ID NO:71, and CDR3 comprises the amino acid sequence of SEQ ID NO:72.

17. The kit of claim 9, wherein the first single-chain antibody and the second single-chain antibody dimerize in the presence of cannabidiol.

18. A method for detecting cannabidiol, comprising contacting a sample suspected of containing cannabidiol with the first single-chain antibody and the second single-chain antibody of the kit of claim 17, and detecting a dimer of the first single-chain antibody and the second single-chain antibody, thereby detecting cannabidiol in the sample.
